

AMENDED CLINICAL TRIAL PROTOCOL 4

COMPOUND: AFLIBERCEPT (AVE0005 / VEGF Trap)

A Multinational, Randomized, Double-Blind Study Comparing Aflibercept Versus Placebo in Patients Treated with Second-Line Docetaxel after Failure of One Platinum Based Therapy for Locally Advanced or Metastatic Non-Small-Cell Lung Cancer (NSCLC)

STUDY NUMBER: EFC10261

STUDY NAME: VITAL

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CLINICAL TRIAL SUMMARY

COMPOUND: Aflibercept

STUDY No : EFC10261

TITLE	A multinational, randomized, double-blind study comparing aflibercept versus placebo in patients treated with second-line docetaxel after failure of one platinum based therapy for locally advanced or metastatic non-small-cell lung cancer (NSCLC).
INVESTIGATOR/TRIAL LOCATION	Multicenter / multinational worldwide except Japan
STUDY OBJECTIVE (S)	<u>Primary objective:</u> To demonstrate overall survival (OS) improvement for aflibercept + docetaxel compared to docetaxel + placebo as second line treatment for patients with locally advanced or metastatic non-small cell lung cancer (NSCLC). <u>Secondary objectives:</u> - To compare efficacy of aflibercept to placebo for: • Progression Free Survival (PFS), • Response Rate (RR) as per RECIST criteria (JNCI 2000), • Health Related Quality of Life (HRQL) assessed by the Lung cancer symptom scale (LCSS) questionnaire - To assess the overall safety of the two treatment arms. - To assess the pharmacokinetics of intravenous (IV) aflibercept in this patient population. - To determine immunogenicity of IV aflibercept (anti- aflibercept antibody detection) in all patients.
STUDY DESIGN	This is a prospective, multicenter, multinational, randomized (1:1), double-blind, parallel-group study comparing the efficacy in term of overall survival (OS) of aflibercept (Arm A) versus placebo (Arm B) administered on top of the standard docetaxel regimen as second-line treatment for patients with NSCLC. Each patient will be treated until disease progression, unacceptable toxicity, or patient's refusal. An executive Steering Committee will be responsible for supervising the progress of the trial. This committee will include the Study Chairman, designated main Investigators and the Sponsor's representatives. An independent Data Monitoring Committee (DMC) will periodically assess the patients' safety data, risk / benefit to the patients and will advise the executive Steering Committee and Sponsor regarding the continued safety of treating ongoing and future study patients. The DMC will also be in charge of reviewing the formal statistical interim analysis of OS planned during the course of the study (see statistical considerations). Allocation to the two treatment arms will be done centrally via an Interactive Voice Response System (IVRS), using a permuted-block randomization according to the following factors: • Eastern Cooperative Oncology Group (ECOG) performance status (0 vs. 1 vs. 2) • Prior or not bevacizumab

STUDY POPULATION	
Main selection criteria:	<u>Inclusion criteria:</u> • Histological/cytological proven locally advanced or metastatic non-small cell lung cancer. • Disease progression during or after one, and only one, prior anticancer therapy which is platinum-based (chemotherapy or targeted therapy) for advanced or metastatic disease. Disease progression within 6 months of adjuvant platinum-based chemotherapy is accepted. <u>Exclusion criteria:</u> - Related to the methodology • Squamous histology/cytology • Less than 28 days elapsed from prior treatment with radiotherapy, surgery, or chemotherapy to the time of randomization. Less than 42 days elapsed from prior major surgery (lung resection) to the time of randomization. • Prior isotope therapy, whole pelvic radiotherapy, or radiotherapy to > 25% of bone marrow • Adverse events (excluding alopecia and those listed in the specific exclusion criteria) from any prior anticancer therapy of grade >1 (National Cancer Institute Common Terminology Criteria [NCI CTCAE] v.3.0) at the time of randomization. • Age <18 years. • Eastern Cooperative Oncology Group (ECOG) performance status (PS) > 2. • History of brain metastases, uncontrolled spinal cord compression, or carcinomatous meningitis, or new evidence of brain or leptomeningeal disease. • History of another neoplasm. Adequately treated basal cell or squamous cell skin cancer, or in situ cervical cancer, or any other cancer from which the patient has been disease-free for > 5 years are allowed. • Participation in another clinical trial and any concurrent treatment with any investigational drug within 30 days prior to randomization. • Any of the following events within the 3 months prior to randomization: treatment resistant peptic ulcer disease, erosive oesophagitis or gastritis, grade 3 or 4 gastrointestinal bleeding/hemorrhage, infectious or inflammatory bowel disease, diverticulitis, pulmonary embolism, or other uncontrolled thromboembolic event. • Any of the following events within the 6 months prior to randomization: myocardial infarction, severe/unstable angina pectoris, coronary/peripheral artery bypass graft surgery, NYHA class III or IV congestive heart failure, stroke or transient ischemic attack. • Occurrence of deep vein thrombosis within 4 weeks, prior to randomization. • Acquired immunodeficiency syndrome (AIDS-related illnesses)

	or known human immunodeficiency virus (HIV) disease requiring antiretroviral treatment - Any severe acute or chronic medical condition, which could impair the ability of the patient to participate to the study or interfere with interpretation of study results. - Absence of signed and dated Institutional Review Board (IRB)-approved patient informed consent form prior to enrollment into the study. - Pregnant or breast-feeding woman. Positive serum or urine pregnancy test prior to randomization. - Patient with reproductive potential (M/F) who do not agree to use accepted and effective method of contraception during the study treatment period and for at least 6 months after the completion of the study treatment. The definition of "effective method of contraception" will be based on the investigator's judgment. For female patients enrolled in the United Kingdom (unless surgically sterile, post-menopausal or for another reason have no chance of becoming pregnant) the following forms of contraception are acceptable: oral contraceptives accompanied by the use of a second method of contraception, as it is not known how oral contraceptives interact with all the study medications or Intra Uterine Device (IUD). For male patients their partner should use an effective means of contraception as described above. - Related to aflibercept - History of prior discontinuation of any anti-VEGF agent due to adverse drug reaction. - Urine protein:creatinine ratio (UPCR) > 1 on morning spot urinalysis or proteinuria > 500 mg/24h. - Serum Creatinine > 1.5 x ULN (if creatinine 1.0 - 1.5 x ULN, creatinine clearance calculated according to Cockcroft-Gault formula < 60 mL/min will exclude the patient). - Uncontrolled hypertension, defined as blood pressure >150/100 mm Hg (grade ≥ 2 according to NCI CTCAE v.3.0), or systolic blood pressure >180 mm Hg if diastolic blood pressure <90 mm Hg, on at least 2 repeated determinations on separate days, within 3 months prior to study randomization. - Patients on anticoagulant therapy with unstable dose of warfarin and/or having an out-of-therapeutic range INR (>3) within 4 weeks prior to randomization. - Evidence of clinically significant bleeding diathesis including hemoptysis, or underlying coagulopathy (e.g. INR>1.5 without vitamin K antagonist therapy), non-healing wound. - Related to docetaxel - Prior docetaxel treatment. - Patients refractory to first line paclitaxel-based therapy (patients are considered refractory if their best response on treatment was Progressive Disease). - History of hypersensitivity to docetaxel, or polysorbate 80.
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	• Inadequate organ and bone marrow function as evidenced by: – Hemoglobin < 10.0 g/dL – Absolute neutrophil count < 1.5 x 10⁹/L – Platelet count < 100 x 10⁹/L – AST/SGOT and/or ALT/SGPT > 2.5 x ULN – Alkaline phosphatase (AP) > 5 x ULN – AST/SGOT and/or ALT/SGPT > 1.5 x ULN concomitant with AP > 2.5 X ULN – Total bilirubin > 1.0 x ULN • Contraindications to the use of corticosteroid treatment. • Symptomatic peripheral neuropathy grade > 2 (National Cancer Institute Common Terminology Criteria [NCI CTCAE] v.3.0).
Total expected number of patients:	Approximately 900 patients (i.e. 450 patients per arm).
Expected number of sites:	Approximately 120 sites are planned.
INVESTIGATIONAL PRODUCT(S) Formulation(s): Route(s) of administration: Dose regimen:	• Aflibercept (also referred as VEGF Trap or AVE0005 in some sections of the protocol) : in 5 mM phosphate, 5 mM citrate, 100 mM sodium chloride, 20% (w/v) sucrose, and 0.1% (w/v) polysorbate 20, pH 6.0, supplied in sealed, sterile, single-use 5-mL vials filled with 4.4 mL with a withdrawable content of 4.0 mL at a concentration of 25 mg/mL. The content of the vial must be diluted prior to infusion with 0.9% sodium chloride or 5% dextrose solution. • Placebo for aflibercept: Sterile aqueous buffered vehicle pH 6.0, containing 5 mM phosphate, 5 mM citrate, 100 mM sodium chloride, 0.1% (w/v) polysorbate 20 and 20% (w/v) sucrose, supplied in sealed, sterile, single-use 5-mL vials filled with 4.4 mL with a withdrawable content of 4.0 mL at a concentration of 25 mg/mL. The content of the vial must be diluted prior to infusion with 0.9% sodium chloride or 5% dextrose solution. • Docetaxel: Marketed formulations will be used for docetaxel: see manufacturer's product information. Aflibercept, placebo and docetaxel will be administered by IV route. Patients will be randomly assigned to receive either Arm A or Arm B: Arm A: <u>Aflibercept</u> 6 mg/kg over 1 hour IV on Day 1 every 3 weeks. or Arm B: <u>Placebo</u> 6 mg/kg over 1 hour IV on Day 1 every 3 weeks. For Arm A and Arm B, immediately followed by: <u>Docetaxel</u> 75 mg/m² IV over 1 hour, on Day 1. • Pre- and post-medication for docetaxel: dexamethasone 8 mg PO (or equivalent) will be administered, evening before Day 1, on Day 1 (early morning, 1 hour before docetaxel treatment, and evening) and on Day 2 (morning and evening). In case of Body Surface Area (BSA) > 2.2 m², the actual dose of

	docetaxel should be adjusted to a maximum BSA of 2.2 m² for safety reasons. Dose reduction and/or treatment delay and/or treatment discontinuation are planned in case of severe toxicity. A cycle (C) is defined as a 3 week-period.
PRIMARY ENDPOINT (S) AND MAIN SECONDARY ENDPOINT (S)	<u>Primary end-point:</u> Overall survival (OS) will be defined as the time interval from the date of randomization to the date of death due to any cause. <u>Secondary End-points:</u> - Efficacy: • PFS: will be defined as the time interval between the date of randomization and the date of tumor progression or death of any cause, whichever comes first. • RR: objective responses include complete responses (CR) and partial responses (PR), as assessed by investigators according to RECIST criteria (for patients with measurable disease at study entry). Confirmation of objective responses will be performed by repeat tumor imaging (CT scan, MRI) at least 4 weeks after the first radiological documentation of response. • Health-related quality of life (LCSS): questionnaires will be self-administered by the patients. Visual analogue scales with 100-mm lines are used to measure the intensity of patient responses. The LCSS total score is defined as the mean of the 9 items of the scale (score between 0 to 100). In addition, the Average Symptom Burden Index (ASBI) will be calculated as the mean of the 6 major lung cancer symptoms. Each individual score, the ASBI and the total scores will be described at each assessment as well as change from baseline scores. - Safety profile of the study treatment in term of AEs/ SAE's and laboratory parameters: • Type according to MedDRA (Medical Dictionary for Regulatory Activities), frequency, severity according to NCI CTCAE v.3.0, seriousness, and relatedness of study treatment-emergent adverse events will be assessed. • Laboratory abnormalities will be assessed according to the NCI CTCAE v.3.0. - Pharmacokinetic parameters and immunogenicity : • In all treated patients, blood sampling will be performed at specified time points during the study, in both arms in order to detect anti-aflibercept antibodies (samples for pharmacokinetics parameters will be taken in parallel)
ASSESSMENT SCHEDULE	- Clinical examinations (including blood pressure, ECOG performance status), laboratory safety tests (including complete blood counts, serum chemistry and urinalysis), and signs and symptoms/adverse events will be obtained prior to drug administration, and every visit before treatment and up to 30 days after the last study treatment administration (data will be collected until the study cut-off date). Blood pressure will be measured once

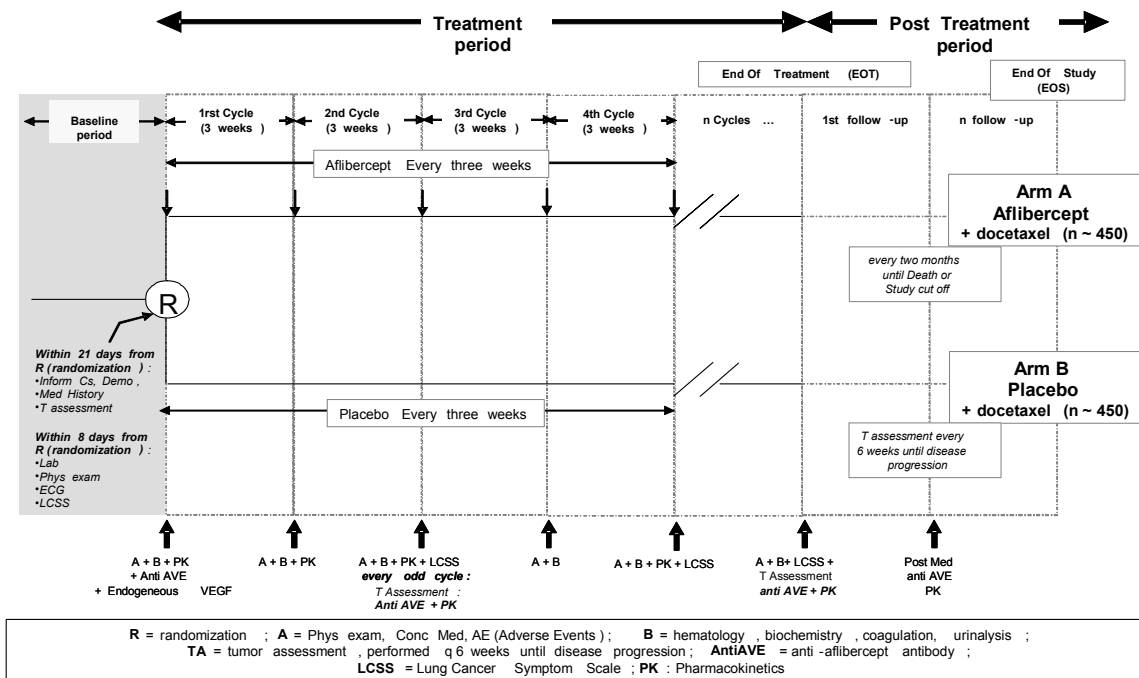
	between cycles. Start collection of Serious Adverse Events from the time of informed consent signature. - LCSS questionnaire will be completed 4 times during the study: at baseline, at cycles 2 and 4, and at the end of treatment. - For all patients, samples for detecting anti-aflibercept antibodies will be collected at baseline, then every other cycle before study drug administration (i.e. samples collected pre-dose on C3, C5, and so on) and then approximately 30 days and finally approximately 90 days after the last administered dose of aflibercept or its placebo. Samples for antibody detection should also be collected in patients who develop specific adverse events. - For all patients, pharmacokinetic blood samples will be collected at the same above time points and, in addition, at the end of aflibercept/placebo infusion at C1. In all patients from selected centers endogenous VEGF will be measured at baseline. - Tumor assessments (chest, brain and abdominal computerized tomography (CT) or magnetic resonance imaging (MRI) scans, except bone scan, and any other exams as clinically indicated), using the same method used at baseline, will be performed every 6 weeks (2 cycles) until disease progression, even if the patient discontinued study treatment before disease progression, or study cut-off date whichever comes first. Confirmatory radiological evaluation will be performed 4 to 6 weeks after initial documentation of response. For all patients, bone scan will be performed at baseline, and then: i) for patients with no bone lesions at baseline, whenever new or worsening bone symptoms occur, ii) for patients presenting bone lesions, every 3 months during study treatment (or at any time if there is a clinical suspicion of bone progression), at the end of treatment and then every 3 months until disease progression is documented. Confirmatory bone scan evaluation will be performed 4 to 6 weeks after initial documentation of response. - After disease progression, patients will be followed every 2 months until death or withdrawal of patient consent. Details of any further anticancer treatment will be collected.
STATISTICAL CONSIDERATIONS	Sample size justification: The cut-off date for the OS and secondary endpoints analysis will be the date when the required number of deaths has been observed. Randomization: Treatment allocation will be performed through IVRS using a pre-

	defined randomization scheme, within 3 days from first dosing. Stratification: Stratification factors will be: ECOG PS (0 vs. 1 vs. 2), prior or not bevacizumab. Analysis populations: • Intent-to-Treat (ITT) population: This population includes all randomized patients. This population is the primary population for all efficacy parameters. All analyses using this population will be based on the treatment assigned by IVRS. • Evaluable Patient (EP) population for tumor response: this population consists of all randomized patients with measurable disease at study entry who have received at least one full cycle of treatment, with lung cancer previously treated with a platinum based chemotherapy, and with at least one post baseline tumor evaluation, except for early disease progression/death. The evaluable patient population for tumor response will be used for the RR analysis (secondary population for RR analysis). All analyses using this population will be based on the treatment actually received. • All-Treated (AT) population : The AT population is a subset of the ITT population that took at least one dose of study medication (aflibercept, placebo or docetaxel), used for safety analysis. All analyses using this population will be based on the treatment actually received. Primary Analysis of the primary endpoint OS is defined as the number of months from the date of randomization to the date of death. If death is not observed during the study, data on OS will be censored at the last date when patient is known to be alive or the cut-off date. OS will be compared between the two treatments by the log-rank test procedure stratified by stratification factors specified at the time of randomization. The estimates of the hazard ratio and corresponding 95% confidence interval will be provided using a Cox proportional hazard model stratified by the same stratification factors as those used for the log-rank test described above. The survival curves will be estimated using Kaplan-Meier estimates. Interim and final analyses One interim analysis of OS is planned for the purpose of futility and overwhelming efficacy when approximately 344 OS events (50% information fraction) have occurred. Calculations are based on an O'Brien-Fleming α-spending function for efficacy boundaries and a gamma (-6) spending function for futility boundaries and assuming that the interim look will actually be carried out after the 344th event. The one-sided nominal significance level to consider study termination for overwhelming efficacy is 0.0015 at the interim analysis. Futility will be considered if the hazard ratio is ≥ 1.03 in favor of the placebo arm. The interim analysis will be carried out under the supervision of the DMC. The final OS analysis will be conducted when 687 deaths have
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	been observed. The one-sided nominal significance level to declare superiority of aflibercept at the final analysis is 0.0245.
DURATION OF STUDY PERIOD (per Patient)	Patients will be treated until progressive disease, unacceptable toxicity, patient's refusal of further study treatment. All patients will be followed when on study treatment and after completion of study treatment during follow up period until death or the study cutoff date, whichever comes first. The maximum study duration and the final study cutoff date for survival will be the date when approximately 687 deaths have occurred.

1 FLOW CHARTS

1.1 GRAPHICAL STUDY DESIGN



1.2 STUDY FLOWCHART

Evaluation	Baseline period	Study phase ^(a)			
		Treatment Period			Post treatment Follow-up Until Death
DAY	Within 21 days before randomization	Every cycle	Every 2 cycles	End of Treatment /withdrawal	
Baseline documentation:					
Informed Consent ^[b]	Before any study protocol specific procedure				
Inclusion/Exclusion Criteria	Within 8 days				
Patient Demography	X				
Prior Medical, Surgical and Oncologic History ^[c]	X				
Physical Examination / ECOG Performance Status ^[d]	Within 8 days	X		X	X until Progressive disease
Laboratories Studies:					
Hematology ^[e] Coagulation ^[f]	Within 8 days	X		X	
Blood and urine Chemistry ^[g]	Within 8 days	X		X	
Pregnancy test ^[h]	Within 8 days				
12 lead Electrocardiogram ^[i]	Within 8 days				
Randomization ^[j]		Cycle 1 within 3 days after randomization			
Study Drug Administration ^[k]		X			
Other Clinical Assessments:					
Adverse Events Assessment ^[l]	X	X		X	X
Prior/Concomitant/Post Medications ^[m]	X	X		X	X
Other investigations	As clinically indicated	As clinically indicated			
Efficacy:					
Tumor assessment ^[n]	X		X	X	X until Progressive disease
Health related Quality of life – LCSS questionnaire ^[o]	Within 8 days		Cycles 2 & 4	X	
Survival status ^[p]		X			X
Samples for Pharmacokinetics ^[q]	X	Cycle 1 at peak & through	X	X	90 days
Serum for Anti-aflibercept Antibody & VEGF level ^[r]	X		X	X	90 days

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- a. **Study Phase:** All assessments should be performed prior to dosing with study drug unless otherwise indicated. Acceptable time windows for performing the assessments are noted below each scheduled treatment day in the Study Flowchart, unless otherwise noted in the footnotes below. Time between biological work-up and physical exam or at baseline and first infusion should not exceed 8 days. If time exceeds 8 days, biological tests, clinical benefit and physical exam must be repeated.
- b. Prior to any study procedures.
- c. **Medical, Surgical and Oncologic History:** including cancer diagnosis (primary tumor characteristics and metastatic sites), prior surgery, radiotherapy, and systemic anticancer therapy and concurrent illness.
- d. **Physical Examination:** Examination of major body systems, blood pressure (BP will be closely monitored for adjustment in antihypertensive therapy as needed in between scheduled treatment visit; in addition BP will be measured during week 2 of each cycle), height (Screening only), body weight, and ECOG PS.
- e. **Hematology:** Hemoglobin, WBC, ANC, platelet count prior to day 1 of every cycle. This blood count will be performed within the cycle in case of fever or infection (see Section 10).
- f. **Coagulation:** prothrombin time, expressed as international normalized ratio (INR), will be assessed in patients under Vitamin K antagonist therapy.
- g. **Blood and urine Chemistry:** Sodium, potassium, calcium, phosphorus, BUN, magnesium, creatinine, total protein, albumin, SGOT (AST), SGPT (ALT), total bilirubin, alkaline phosphatase (AP), and glucose. **Urinalysis:** dipstick (WBC, RBC); Urinary protein and urinary creatinine and UPCr will be calculated on morning urine spot prior to dosing of each cycle. During study treatment, if proteinuria from renal origin (according to urine protein electrophoresis) is associated with hematuria then a clinical work up to rule out thrombotic microangiopathy should be performed: it may include LDH, haptoglobin, schistocytes and osmotic fragility (if available) measured in blood.
- h. **Pregnancy Test:** Women of reproductive potential must have a negative pregnancy test result (serum or urine) within 8 days prior to first infusion. To be repeated as clinically indicated.
- i. **12 lead ECG:** To be repeated as clinically indicated.
- j. **Randomization:** All eligible patients will be randomly assigned to one of the two treatment groups (either aflibercept or Placebo) using an Interactive Voice Response System (IVRS). Study treatment should be started within 3 days from randomization.
- k. **Study Drug Administration:** Aflibercept or Placebo will be administered every 3 weeks immediately followed by docetaxel administered on Day 1 of each 21-day cycle. Cycle administration will start within 3 days of randomization, and then repeated every 3 weeks (see Section 8.4).
- l. **Adverse Event Assessment:** The period of safety observation starts from the time the patient gives informed consent. All signs and symptoms observed from the time of signature of informed consent will be recorded as adverse events only if they are still present at the time of first study drug administration (and reported in cycle 1 of the CRF) or at any time if they are serious. All adverse events will be recorded until 30 days after the final dose of study drugs. During the follow-up period, only ongoing related or new related adverse events will be recorded. Serious adverse events ongoing at the end of the study treatment will be followed during the follow-up period until resolution or stabilisation regardless relationship with study drugs.
- m. **Assessment of Concomitant Medications:** Concomitant medications and treatments will be recorded from 21 days prior to the start of study drug, before every cycle during the study treatment period, and up to 30 days after the final dose of study drug. Once the patient has withdrawn from study treatment, concomitant medication should only be recorded if used to treat new or unresolved study treatment related adverse events.
- n. **Tumor Assessment:** Chest, brain (at screening only) and abdominal CT or MRI scans and any other exams as clinically indicated to assess target and non target lesions to be performed at baseline (screening), at the end of each even-numbered treatment cycle (i.e. cycle 2, 4, 6, 8, etc), whenever disease progression (according to RECIST) is suspected, and at the end of treatment visit, using the same method for each assessment. For the purpose of analysis, a CR or PR will be considered a confirmed response if a subsequent assessment has been performed 4 – 6 weeks after the first assessment and the results confirm the initial finding. During the follow-up period, patients who went off study treatment prior to documented disease progression will be evaluated by CT-Scan/MRI for tumor progression every 6 weeks from End of Study treatment until disease progression or death. For all patients, a bone scan will be performed at baseline, and then: i) for patients with no bone lesions at baseline whenever new or worsening bone symptoms occur; ii) for patients presenting bone lesions, every 3 months during study treatment (or at any time if there is a clinical suspicion of bone progression), at the end of treatment and then every 3 months until disease progression is documented.
- o. **Health related Quality of life:** At baseline, cycles 2 & 4 and at end of treatment; Lung Cancer symptom scale (LCSS) questionnaire to be self filled in by the patient at the center prior to informing the patient about disease evolution and before the next study treatment. It is mandatory that a key person (e.g. research nurse) at each center be responsible for questionnaire data collection, in order to optimize compliance of the patient and to ensure completeness of the data.

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- p. **Survival Status:** During the follow-up period, once disease progression is documented or other anticancer therapy is started, patients will be followed to collect his/her survival status (dead, alive, or lost to follow-up) until death or cut-off date.
- q. **Pharmacokinetics:** blood sample for measurement of free aflibercept and VEGF:aflibercept complex to be collected: for all patients: pre-dose on cycle 1, at the end of infusion on cycle 1, then pre-dose every other cycle (i.e. samples collected pre-dose on C3, C5, and so on) and then approximately 30 days then 30 days and 90 days after the last aflibercept/placebo administration. In selected centers blood sample for measurement of VEGF endogeneous level will be collected at baseline only.
- r. For all patients, serum for detection of **anti-aflibercept antibodies** will be collected pre-dose on cycle 1, then pre-dose every other cycle, then 30 and finally 90 days after the last aflibercept administration. In addition, serum for detection of anti-aflibercept antibodies will be collected in specific circumstances.

Abbreviations: AE, Adverse Event; ALT, Alanine Aminotransferase; AP, Alkaline Phosphatase; AST, Aspartate Aminotransferase; ANC, Absolute Neutrophil Count; BP, Blood Pressure; BUN, Blood Urea Nitrogen; CR, Complete Response; CT, Computed Tomography; D, day; ECG, Electrocardiogram; ECOG, Eastern Cooperative Oncology Group; INR, International Normalized Ratio; IVRS, Interactive Voice Response System; LCSS, Lung Cancer Symptom Scale; LDH, Lactico Dehydrogenase; MRI, Magnetic Resonance Imaging; PR, Partial Response; PS, Performance Status; RBC, Red Blood Cells, RECIST, Response Evaluation Criteria in Solid Tumors; SGOT, Serum Glutamate-Oxalate Transferase; SGPT, Serum Glutamate-Pyruvate Transferase; WBC: White Blood Cell, UPCR urinary protein creatinine ratio

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3 LIST OF ABBREVIATIONS

Ab	Antibodies
AE	Adverse Event
AIDS	Acquired Immunodeficiency Syndrome
ALT (SGPT)	Alanine Aminotransferase (or Serum Glutamate-Pyruvate Transferase)
ANC	Absolute Neutrophil Count
AP	Alkaline phosphatase
AST (SGOT)	Aspartate Aminotransferase (or Serum Glutamate-Oxaloacetate Transferase)
ASBI	Average Symptom Burden Index
AT	All-Treated
AUC	Area Under the Curve
BP	Blood Pressure
BSA	Body Surface Area
BSC	Best Supportive Care
BUN	Blood Urea Nitrogen
CBC	Complete Blood Count
CHO	Chinese Hamster Ovary
CI	Confidence Interval
CNS	Central Nervous System
COPD	Chronic Obstructive Pulmonary Disease
CRF	Case Report Form
CSR	Clinical Study Report
CT	Computerized Tomography
CTCAE	Common Terminology Criteria for Adverse Events
D	Day
DEHP	Di (2-ethylhexyl) phthalate
DLT	Dose Limiting Toxicity
DMC	Data Monitoring Committee
DRF	Discrepancy Resolution Form
DVT	Deep Vein Thrombosis
EC	Ethics Committee
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
EGFR	Epidermal Growth Factor Receptor
ELISA	Enzyme Linked ImmunoSorbent Assay
EMA	European Medicine Evaluation Agency
EP	Evaluable Patient
FDA	Food and Drug Administration
FLAIR	Fast Fluid-Attenuated Inversion-Recovery
GCP	Good Clinical Practice
G-CSF	Granulocyte-colony-stimulating factor
HA	Health Authorities
HBP	High Blood Pressure

HIV	Human Immunodeficiency Virus
HR	Hazard Ratio
HRQL	Health Related Quality of Life
HSR	Hypersensitive reaction
IA	Interim Analysis
IB	Investigators' Brochure
ICH	International Conference on Harmonization
Ig	Immunoglobulin
IND	Investigational New Drug
INR	International Normalized Ratio
IP	Investigational Product
IRB/IEC	Institutional Review Board/Independent Ethics Committee
ITT	Intent-to-Treat
IUD	Intra Uterine Device
IV	Intravenous
IVRS	Interactive Voice Response System
LCSS	Lung cancer symptom scale
LD	Longest diameter
LDH	Lactico dehydrogenase
MedDRA	Medical Dictionary for Regulatory Activities
MRI	Magnetic Resonance Imaging
NCI	National Cancer Institute
NSCLA	Non-Small-Cell Lung Adenocarcinoma
NSCLC	Non-Small-Cell Lung Cancer
NYHA	New York Heart Association
OS	Overall Survival
PD	Progressive Disease
PE	Pulmonary Embolism
PI	Package Insert
PFS	Progression-Free Survival
PO	Per Oral Route
PPI	Present Pain Intensity
PS	Performance Status
PT	Prothrombin Time
Q	Every
PVC	Polyvinyl Chloride
RBC	Red Blood Cell
RECIST	Response Evaluation Criteria In Solid Tumors
RPLS	Reversible Posterior Leukoencephalopathy Syndrome
RR; OR, CR, PR	Response Rate; Overall, Complete, Partial Response
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SC	Subcutaneous
SD	Standard deviation
T	Temperature
TA	Tumor Assessment

TEAE	Treatment-Emergent Adverse Event
TMA	Thrombotic microangiopathy
TOTM	Tri-2-Ethylhexyl Trimellitate
TXT	Taxotere ®
ULN	Upper Normal Limit
UPCR	Urinary Protein:Creatinine Ratio
US	United States/Ultrasounds
VEGF	Vascular Endothelial Growth Factor
VEGFR	Vascular Endothelial Growth Factor Receptor
W	Week
WBC	White Blood Cells

4 INTRODUCTION AND RATIONALE

4.1 NON-SMALL CELL LUNG CANCER

Lung cancer is the leading cause of cancer death worldwide. It is estimated that approximately 138,000 new cases of non-small-cell lung cancer (NSCLC) were diagnosed in the USA in 2004. Most patients have advanced disease at diagnosis (Stages IIIB and IV) for which there is presently no cure.

NSCLC includes several histological subtypes. Non-Small-Cell Lung Adenocarcinoma (NSCLA) accounts for approximately 50% of all cases of NSCLC, and its incidence continues to increase. While chemotherapy is used in all histological subtypes, the newer targeted therapies are more likely to benefit patients with specific clinical, pathologic, and molecular profiles. Patients with epidermal growth factor receptor (EGFR) mutations, women, asian patients, and nonsmokers have higher probabilities of response to the EGFR tyrosine kinase inhibitors erlotinib and gefitinib [1, 2]. Adenocarcinoma, particularly the bronchoalveolar subtype, appears to be more sensitive to antagonists of EGFR [3, 4, 5].

Current therapies for advanced disease prolong survival and improve disease-related quality of life in well-defined patient populations. In patients ≤ 65 years and ECOG performance status (PS) 0 or 1, initial therapy with double-combination platinum-based chemotherapy has shown to improve survival [6] while in patients older than 65 years, monotherapy is better tolerated and has also been shown to have a survival effect [7, 8]. Several double combinations with approved agents are commonly used and have shown similar efficacy [9]; although, some data suggest superiority of docetaxel-based regimens [10]. Objective responses range from 25% to 35%, median survival of 8 to 15 months, and 1-year survival rates of 25% to 35%. A small fraction of patients with ECOG PS 2 have been treated in most Phase III chemotherapy trials, but in general, they do not tolerate chemotherapy as well. However, these patients have benefited from the newer targeted-specific therapies that appear to be better tolerated than chemotherapy.

In patients with advanced NSCLC whose disease becomes resistant to platinum-containing regimen, administration of docetaxel results in longer survival than either best supportive care alone [11] or other chemotherapy [12]. The antifolate pemetrexed (Alimta®), and docetaxel (Taxotere®) have shown equivalent activity in the second-line treatment setting: median PFS was 2.9 months for each arm and median survival time was 8.3 versus 7.9 months (P = not significant) for pemetrexed and docetaxel, respectively [13]. Both drugs have received FDA and EMEA approval for the treatment of platinum-resistant NSCLC.

Two molecularly targeted drugs that inhibit the EGFR have also shown to be effective in the platinum-resistant advanced NSCLC treatment setting. Erlotinib (Tarceva®) has demonstrated an improvement in survival and in tumor-related symptoms compared to best supportive care alone [14] whereas gefitinib (Iressa®) induced objective responses and symptoms improvement [15, 16] though has not demonstrated a survival benefit [17].

However, globally, the results of presently available regimens remain modest and new drugs for the treatment of those patients are needed.

4.2 ANGIOGENESIS INHIBITION IN NSCLC

Overexpression of VEGF/VEGFR is common in lung cancer, and there is a close correlation between the expression of VEGF/VEGFR, degree of neoangiogenesis, and survival [18], especially in patients with NSCLA [19, 20].

It was previously demonstrated that the addition of angiogenesis inhibitors to standard anticancer drugs can improve the efficacy of chemotherapy in the treatment of patients with advanced NSCLA. The evaluation of anti-VEGF agents in the treatment of patients with advanced stage NSCLC is presently limited to patients with non-squamous NSCLC, or NSCLA, without central nervous system (CNS) metastases due to higher risk of life-threatening or fatal hemorrhages [21, 22].

4.3 INVESTIGATIONAL PRODUCT

Aflibercept (known as AVE0005 or VEGF Trap) is a recombinantly-produced fusion protein consisting of human VEGF receptor extracellular domains fused to the F_c portion of human IgG₁. It contains sequences encoding Ig domain 2 from VEGFR1 fused to Ig domain 3 from VEGFR2, which in turn is fused to the hinge region of the human IgG₁ F_c domain. Aflibercept is a dimeric glycoprotein with a molecular weight of 115 kDa. Aflibercept is made recombinantly in Chinese hamster ovary (CHO) cell.

Aflibercept inhibition of tumor growth in mouse xenograft models was observed at ≥ 2.5 mg/kg twice weekly dose, which corresponded to a pharmacological exposure where free aflibercept increases approximately linearly in excess of plateauing VEGF:aflibercept complex (i.e., bound aflibercept) levels.

Preclinical data has shown that aflibercept is synergistic with docetaxel. Detailed preclinical data are described in the Investigators' Brochure [23].

4.4 SUMMARY OF PREVIOUS HUMAN EXPERIENCE

Detailed information regarding the clinical safety, pharmacokinetics, pharmacodynamics, and preliminary efficacy of aflibercept can be found in the aflibercept (AVE0005) Investigator's Brochure [23]. So far, around 500 patients have been treated with aflibercept, either in monotherapy or combined with various chemotherapies (oxaliplatin, 5-FU, CPT-11, gemcitabine, docetaxel, docetaxel/cisplatin and docetaxel/cisplatin/5-fluorouracil). In general, aflibercept has an acceptable safety profile at the dose levels evaluated to date, as the majority of adverse events encountered have been mild to moderate in severity. Adverse reactions linked to VEGF blockade have been consistent in incidence, severity, across all the studies.

4.4.1 Summary of experience of aflibercept in monotherapy

4.4.1.1 Clinical results

Study TED6115/6116 is an ongoing open-label dose escalation study of aflibercept given in monotherapy in patients with various advanced cancers. Doses ranging from 0.3 mg/kg to 7 mg/kg i.v, administered once every 2 weeks, were investigated with a median of 3 to 7 cycles. Overall 46 patients have been enrolled and received 1 to 50 cycles.

Six out of 46 (13%) patients experienced dose-limiting toxicity (ies) (DLT) within the first 4 weeks of study treatment, influencing dose escalation or cohort extension decision. One patient at the 1.0 mg/kg dose experienced grade 3 arthralgia associated with grade 3 dysphonia, at the 2.0 mg/kg dose, 1 patient experienced grade 3 dyspnea (patient with preexisting COPD). At 4.0 mg/kg dose, 1 patient experienced grade 3 hypertension and another patient had grade 2 proteinuria (>2.0 g/24hours) associated with grade 3 manageable hypertension. Two patients at the 7.0 mg/kg dose level reported DLTs, 1 grade 3 rectal ulcer and the other patient developed grade 3 proteinuria that recovered and continued treatment at the 4mg/kg dose.

Treatment emergent adverse events, consisting mostly of low grades events, were reported in the most of the patients (Table 1). No evident particular pattern, in frequency or in severity of the adverse events, could be identified across the investigated doses. The 5 most frequent TEAEs reported were fatigue (84.8%), nausea (65.2%), constipation (54.3%), dysphonia (50.0%) and dyspnea (50.0%).

Adverse events that may be related to VEGF blockade, were observed across all dose levels. Overall incidences for these events were: dysphonia (50.0%), hypertension (41.3%), proteinuria (19.6%), epistaxis (13.0%). No arterial ischemic event was reported.

Table 1 - Most common TEAEs All Grades in studies TED6115 and TED6116 (in at least 10% of treated patients), regardless of relationship to study medication

MedDRA Preferred Term	Dose Level							Total Doses (N=46)
	0.3 mg/kg (N=3)	1.0 mg/kg (N=7)	2.0 mg/kg (N=6)	3.0 mg/kg (N=7)	4.0 mg/kg (N=7)	5.0 mg/kg (N=4)	7.0 mg/kg (N=12)	
	N	N	N	N	N	N	N	
Subjects With at Least 1 TEAE	3	7	6	7	7	4	10	44
Subjects Without Any TEAEs	0	0	0	0	0	0	2	2
Fatigue	3	6	6	7	7	2	8	39
Nausea	3	5	2	7	6	2	5	30
Constipation	1	4	6	5	4	1	4	25
Dysphonia	1	2	3	6	3	2	6	23
Dyspnoea	1	4	5	6	3	0	4	23
Vomiting	2	3	3	6	3	2	2	21
Hypertension	0	2	2	1	4	3	7	19
Abdominal pain	2	3	1	4	3	1	3	17
Anorexia	1	3	3	4	4	1	1	17
Arthralgia	1	3	2	3	2	2	4	17
Back pain	1	1	1	4	3	3	4	17
Headache	0	3	3	1	3	3	3	16
Diarrhoea	2	3	1	4	3	0	2	15
Pyrexia	1	3	3	3	2	1	2	15
Shoulder pain	0	3	1	1	3	3	3	14

MedDRA Preferred Term	Dose Level							Total Doses (N=46)
	0.3 mg/kg (N=3)	1.0 mg/kg (N=7)	2.0 mg/kg (N=6)	3.0 mg/kg (N=7)	4.0 mg/kg (N=7)	5.0 mg/kg (N=4)	7.0 mg/kg (N=12)	
	N	N	N	N	N	N	N	
Cough	0	3	3	2	2	0	1	11
Myalgia	0	2	2	2	1	1	3	11
Pain in extremity	0	1	2	2	0	2	4	11
Pharyngolaryngeal pain	0	2	2	2	1	1	2	10

The most frequently reported grade 3-4 AE was HBP, in 15/46 (32.6%) of the patients (Table 2). At doses of 4 mg/kg or above, half of the patients (11 out of 23 treated patients) experienced a blood pressure increase (grade ≥ 3). Grade 3 HBP seems to occur earlier in the treatment as the dose is escalated from 4 mg/kg. Hypertension is reported within 2 weeks of the first drug administration. Severe proteinuria (grade 3) was observed in 2 (4.3%) patients. Fatigue was the only other Grade 3 AE to be reported in more than 5 patients.

Table 2 - Most common TEAEs Grade 3-4 in studies TED6115 and TED6116 (in at least 2 treated patients), regardless of relationship to study medication

MedDRA Preferred Term	Dose Level							Total Doses (N=46)
	0.3 mg/kg (N=3)	1.0 mg/kg (N=7)	2.0 mg/kg (N=6)	3.0 mg/kg (N=7)	4.0 mg/kg (N=7)	5.0 mg/kg (N=4)	7.0 mg/kg (N=12)	
	N	N	N	N	N	N	N	
Subjects With at Least One Grade 3-4-5 TEAE	1	6	4	5	5	4	8	33
Subjects Without Any Grade 3-4-5 TEAEs	2	1	2	2	2	0	4	13
Hypertension	0	1	2	1	3	3	5	15
Fatigue	0	2	1	2	2	0	0	7
Dyspnoea	0	2	2	0	0	0	0	4
Abdominal pain	0	1	0	1	0	0	1	3
Nausea	0	0	0	1	0	2	0	3
Vomiting	0	0	0	1	0	2	0	3
Anorexia	0	0	0	1	0	1	0	2
Arthralgia	0	1	0	0	0	0	1	2
Blood alkaline phosphatase	0	1	0	1	0	0	0	2
Blood alkaline phosphatase increased	0	0	0	1	1	0	0	2
Blood sodium decreased	0	0	0	0	1	0	1	2
Colon cancer metastatic	0	0	0	0	1	0	1	2
Hyperbilirubinaemia	0	0	0	1	0	0	1	2
Pleural effusion	0	1	0	0	0	1	0	2
Proteinuria	0	0	1	0	0	1	0	2

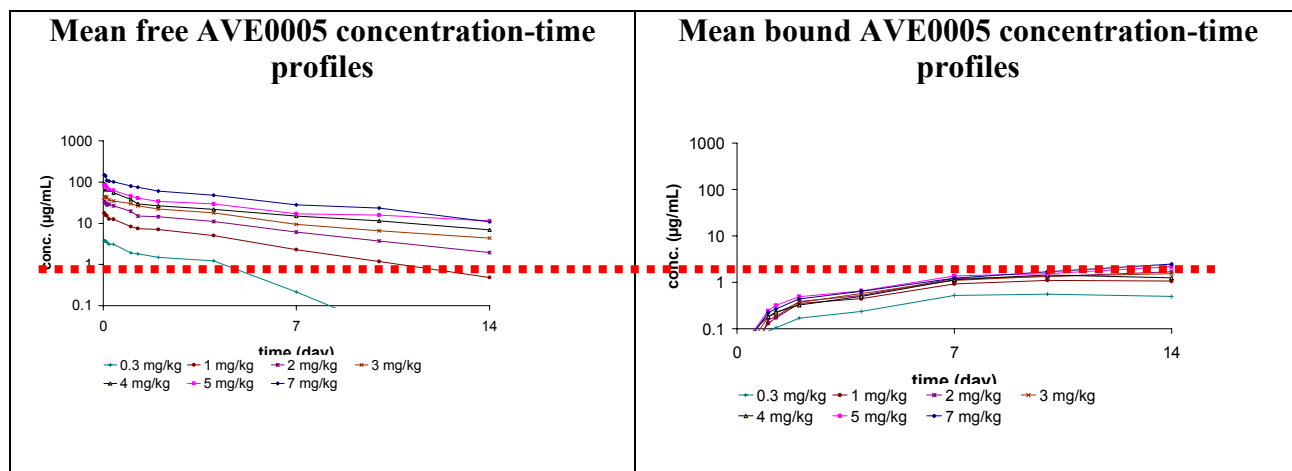
Three patients achieved a partial response. Responses were observed in a patient with malignant thymoma, and 2 other patients with ovarian cancer had tumor responses. All these tumor responses were observed in heavily pretreated patients.

4.4.1.2 Pharmacokinetics results

The pharmacokinetics of free aflibercept appeared to be linear between the 2 and 7 mg/kg dose levels while bound aflibercept increased with dose between 0.3 and 2 mg/kg, then plateaued

between 2 and 7 mg/kg, suggesting that free aflibercept was present in sufficient amount to bind all endogenous VEGF at these higher dose levels. Free aflibercept levels have remained in excess of bound levels (ratio >1) throughout the dosing intervals at ≥ 2.0 mg/kg dose levels (Figure 1).

Figure 1 - Intravenous free versus bound aflibercept concentration time profile (Cycle 1)



4.4.1.3 Conclusion

Based on the above adverse events, thought to be related to VEGF blockade or not, formal DLTs as well as adverse events resulting in discontinuation, safety profile, pharmacokinetic results, and clinical benefit across dose levels, it was recommended to select 4 mg/kg every 2 weeks for aflibercept administered as a single agent.

This is independently supported by the safety experience from currently ongoing phase II studies of aflibercept administered as single agent, in the indications of non-small cell lung adenocarcinoma and advanced ovarian carcinoma (either at doses of 2 or 4 mg/kg every 2 weeks). The safety profile in these trials is similar to the phase I study described above with most adverse events being grade 1 or 2. The adverse events most frequently reported are: hypertension, proteinuria, epistaxis, dysphonia, pain (including abdominal pain and back pain), asthenia/fatigue, nausea, vomiting, headache, constipation, increase in liver enzymes, myalgia, arthralgia, dyspnea, decreased appetite and diarrhea.

4.4.2 Summary of experience of aflibercept in lung cancer

Preliminary safety results

ARD6123 is a Phase II study of the efficacy and safety of aflibercept in patients with platinum- and erlotinib-resistant, locally advanced or metastatic, non-small-cell lung adenocarcinoma (NSCLA). This study was designed as an open-label, multicenter, single-arm, two-stage, Phase II study of aflibercept 4.0 mg/kg IV every 2 weeks in patients with NSCLA.

From the first patient included in February 13, 2006 to December 31, 2006, a total of 56 patients have been enrolled and treated in the study. The database as of November 27, 2006 had 33

patients in the aflibercept IV at 4 mg/kg. This safety summary focuses on the 33 patients who have received aflibercept IV and are part of the database.

There were 15 (46%) of the 33 patients still on study as of the cut-off date, while 18 (55%) of the patients were withdrawn from the study. The main reason for study treatment discontinuation across the 33 patients was disease progression 12 (36%). Two (6%) of the patients withdrew from treatment due to adverse events. Other reasons for study discontinuation include 3 as per patient request and 1 other patient with reason unknown.

A total of 132 cycles of drug were administered across all patients. The minimum number of cycles given was 1; the maximum number was 14. The mean and median number of cycles administered was 4.

Most patients 30/33 (91%) experienced treatment emergent adverse events (TEAEs) during the trial. Most of the events were grade 1 or 2 by NCI criteria, while 13 (39%) patients reported grade 3 or 4 events.

The most common TEAEs experienced (any grade) were: fatigue 18 (55%) in which 2 (6%) patients experienced grade 3 or 4, followed by dysphonia 12 (36%), dyspnea 10 (30%), and anorexia 9 (27%) of the patients. The following AEs: constipation, cough, headache, hypertension, and peripheral edema were all reported in 7 (21%) patients each. Arthralgia, epistaxis, hemoptysis and nausea were reported in 4 (12%) patients each. Decreased appetite, dehydration, dizziness, dysphagia, musculoskeletal pain, pyrexia and urinary tract infection were reported in 3 (9%) patients each.

The overall safety profile of aflibercept at the dose of 4mg/kg in this 3rd line lung cancer study is consistent with the underlying disease. Events of possible cancer baseline etiology include anorexia, anxiety, hemoptysis, decreased appetite, cough, dyspnea, fatigue, vomiting. Possible aflibercept adverse events reported in the trial such as hypertension, headache, epistaxis, proteinuria, fatigue, dyspnea, hemoptysis, stomatitis (mostly grade 1 or 2) are consistent with those reported with other anti-VEGF. Adverse events that seem to be more related to a aflibercept profile include dysphonia seen in this trial in 12 (36%) of patients, and a number of pain related terms that included multiple terms such as musculoskeletal, bone, flank, neck, oral, extremity, jaw, etc. The pain profile of adverse events is mostly consisting of low grades events. The pain profile could also be related to the underlying disease but it seems to be non-specific to tumor sites and needs to be followed to better understand the nature. Therapeutic protein products are known to cause these types of side effects.

Preliminary activity results

Partial response was observed in 2/46 patients, and stable disease (≥ 2 months) was reported in 18/46 patients (investigators assessment, as of January 31st, 2007). Further evaluations of all enrolled patients are ongoing.

4.4.3 Summary of experience of aflibercept in combination with docetaxel

4.4.3.1 Clinical results

TCD6120 (AVE0005A/1004), an ongoing dose-escalation study of aflibercept administered every 3 weeks in combination with docetaxel, then with docetaxel/cisplatin was aimed to determine the recommended dose of aflibercept when combined with docetaxel in advanced solid tumor patients based on safety, tolerability, and pharmacokinetics .

Sequential cohorts of patients have been treated with successively higher doses of aflibercept given with standard doses of docetaxel (75 mg/m²) every 3 weeks, then with standard doses of docetaxel (75 mg/m²) and cisplatin (75 mg/m²) every 3 weeks. The 2.0 mg/kg dose level was the starting dose level of aflibercept with docetaxel, and 4.0 mg/kg was the starting dose of aflibercept with docetaxel/cisplatin.

Until October 30, 2006, a total of 38 patients were enrolled and treated in the escalation step of the study:

- 27 in the aflibercept and docetaxel combination cohort across 6 dose levels: 2mg/kg (7 pts/0 still ongoing), 4 mg/kg (3 pts/1 still ongoing), 5 mg/kg (6 pts/1 still ongoing), 6 mg/kg (4 pts/2 still ongoing), 7 mg/kg (5 pts/1 still ongoing) and 9 mg/kg (2 pts/2 still ongoing).

Three out of 27 aflibercept and docetaxel treated patients experienced a dose limiting toxicity (DLT), one at the 2 mg/kg dose level (neutropenic infection, most probably due to docetaxel-related toxicity), one at the 7 mg/kg dose level (grade 3 dysphonia, knowing that this patient had a grade 1 dysphonia at baseline) and one at the 9 mg/kg dose level (hypertension). Of note, one patient treated at 7 mg/kg experienced an hypertension (HBP), which was less manageable but which was finally controlled with therapy, therefore not meeting DLT criteria.

As of database extract of December 15, 2006, twenty-seven (27/27, 100%) aflibercept and docetaxel treated patients had at least one treatment emergent adverse event (TEAE) (all grades, regardless of relationship to study medication, excluding laboratory adverse events). The most frequent TEAE (all grades, regardless of relationship to study medication) are described in [Table 3](#).

Table 3 - Most common TEAEs in study TCD6120 (all grades, occurring in at least 10% of treated patients), regardless of relationship to study medication

	All doses	
Total number of treated patients [N]	27	
Total patients with any TEAE [N (%)]	27 (100%)	
Total patients with any Grade 3-4 TEAE [N (%)]	17 (67.0%)	
MedDRA SOC term	All Gr	Gr 3
By MedDRA body system		
Eye disorders	12 (44.4)	-
Conjunctivitis	6 (22.2)	-
Lacrimation increased	3 (11.1)	-
Gastrointestinal	24 (88.9)	2 (7.4)
Stomatitis	12 (44.4)	-
Constipation	11 (40.7)	-

	All doses	
Total number of treated patients [N]	27	
Total patients with any TEAE [N (%)]	27 (100%)	
Total patients with any Grade 3-4 TEAE [N (%)]	17 (67.0%)	
MedDRA SOC term	All Gr	Gr 3
By MedDRA body system		
Diarrhoea	11 (40.7)	-
Nausea	8 (29.6)	-
Abdominal pain	7 (25.9)	1 (3.7)
Dyspepsia	4 (14.8)	-
Haemorrhoids	6 (22.2)	-
Odynophagia	3 (11.1)	-
General disorders and administration site conditions	22 (81.5)	8 (29.6)
Fatigue ^a	11 (40.7)	4 (14.8)
Asthenia ^a	10 (37.0)	2 (7.4)
Mucosal inflammation	7 (25.9)	-
Pyrexia	7 (25.9)	-
General physical health deterioration	3 (11.1)	1 (3.7)
Infections and infestations	14 (51.9)	4 (14.8)
Lung Infection	3 (11.1)	1 (3.7)
Investigations	7 (25.9)	-
Weight decrease	7 (25.9)	-
Metabolism and nutrition disorders	7 (25.9)	-
Anorexia	6 (22.2)	-
musculoskeletal and connective tissue disorders	9 (33.3)	-
Bone pain	3 (11.1)	-
Nervous system disorders	14 (51.9)	-
Dysgeusia	8 (29.6)	-
Headache	5 (18.5)	-
Paresthesia	4 (14.8)	-
Neuropathy	2 (7.4)	-
Neuropathy peripheral	1 (3.7)	-
Peripheral sensory neuropathy	1 (3.7)	-
All neuropathy (including the 4 AEs listed above)	8 (29.6)	-
Psychiatric disorder	8 (29.6)	-
Anxiety	5 (18.5)	-
Respiratory, thoracic and mediastinal disorders	27 (100.0)	2 (7.4)
Dysphonia	20 (74.1)	1 (3.7)
Epistaxis	20 (74.1)	-
Dyspnoea	7 (25.9)	1 (3.7)
Rhinorrhoea	6 (22.2)	-
Skin and subcutaneous tissue disorders	21 (77.8)	2 (7.4)
Nail disorder ^b	13 (48.1)	2 (7.4)
Onycholysis ^b	6 (22.2)	-
Alopecia	12 (44.4)	-
Palmar-plantar erythrodysesthesia syndrome	5 (18.5)	2 (7.4)
Vascular disorders	17 (63.0)	2 (7.4)
Hypertension	15 (55.6)	2 (7.4)
Thrombosis	1	-

^a Patient #1503 had both asthenia and fatigue

^b Patient 1601 had both nail disorder and onycholysis

Infections and events of fever in absence of infection include the corresponding neutropenic complications.

Table 4 - Hematological toxicity, worst grade by patient, in study TCD6120

	Initial planned aflibercept dose level (mg/kg) administered every 3 weeks						All doses
	2	4	5	6	7	9	
Total number of treated patients [N]	7	3	6	4	5	2	27
Total number of evaluable patients ^a [N]	7	3	6	4	5	2	27
Leucopenia N (Gr 3-4 N)	7 (7)	3 (3)	6 (5)	4 (4)	4 (2)	2 (2)	26 (23)
Total number of evaluable patients ^a [N]	7	3	6	4	5	2	27
Neutropenia N (Gr 3-4 N)	7 (7)	3 (3)	6 (4)	4 (3)	5 (3)	2 (2)	27 (22)
Total number of evaluable patients ^a [N]	7	3	6	4	5	2	27
Anemia N (Gr 3-4 N)	6 (3)	2 (1)	5 (3)	3 (2)	4 (2)	0 (0)	20 (11)
Total number of evaluable patients ^a [N]	7	3	6	4	5	2	27
Thrombocytopenia N (Gr 3-4 N)	2 (0)	0 (0)	1 (0)	0 (0)	0 (0)	-	3 (0)

^a A patient is evaluable if having at least a blood count for the given test between two infusions.

Patient #2206 is reported as having a false Grade 4 thrombocytopenia due to unit error. These data was not taken into account on this table.

Three confirmed partial response (in breast cancer at dose levels of 2.0; 4.0 and 6.0 mg/kg) and 15 stabilizations (including 2 in lung cancer) were reported.

- 11 patients in the aflibercept and docetaxel/cisplatin combination cohort across 2 dose levels: 4 mg/kg (6 pts/0 still ongoing) and 5 mg/kg (5 pts/3 still ongoing).

One out of 11 aflibercept +docetaxel/cisplatin treated patients experienced a DLT at 4 mg/kg (febrile neutropenia, most probably due to docetaxel-related toxicity). Dose escalation is still ongoing.

4.4.3.2 Pharmacokinetics results

Pharmacokinetics of free and bound aflibercept were determined in patients for which blood samples have been analyzed as of 27 October 2006. The results are summarized in [Table 5](#).

Table 5 - Mean (SD) pharmacokinetic parameters of free and bound aflibercept in TCD6120

Dose (mg/kg)	No of patients	Free aflibercept (cycle 1)			Bound aflibercept	
		Cmax (µg/mL)	Clast (µg/mL)	CL (mL/day/kg)	End of cycle 1	Median of following cycles
2	7	43.2 ± 12.0	0.45 ± 0.58	18.8 ± 5.6	1.61 ± 0.58	2.84 ± 1.06
4	9	106 ± 47	2.22 ± 1.88	13.3 ± 4.5	1.96 ± 0.42	3.44 ± 0.92
5	7	116 ± 47	4.39 ± 2.94	12.6 ± 4.4	2.33 ± 0.47	3.29 ± 1.05
6	4	105 ± 19	3.44 ± 2.14	14.7 ± 5.0	1.93 ± 0.40	2.54 ± 0.38
7	1	182	-*	-*	-*	-*

*: no data available (studies on going)

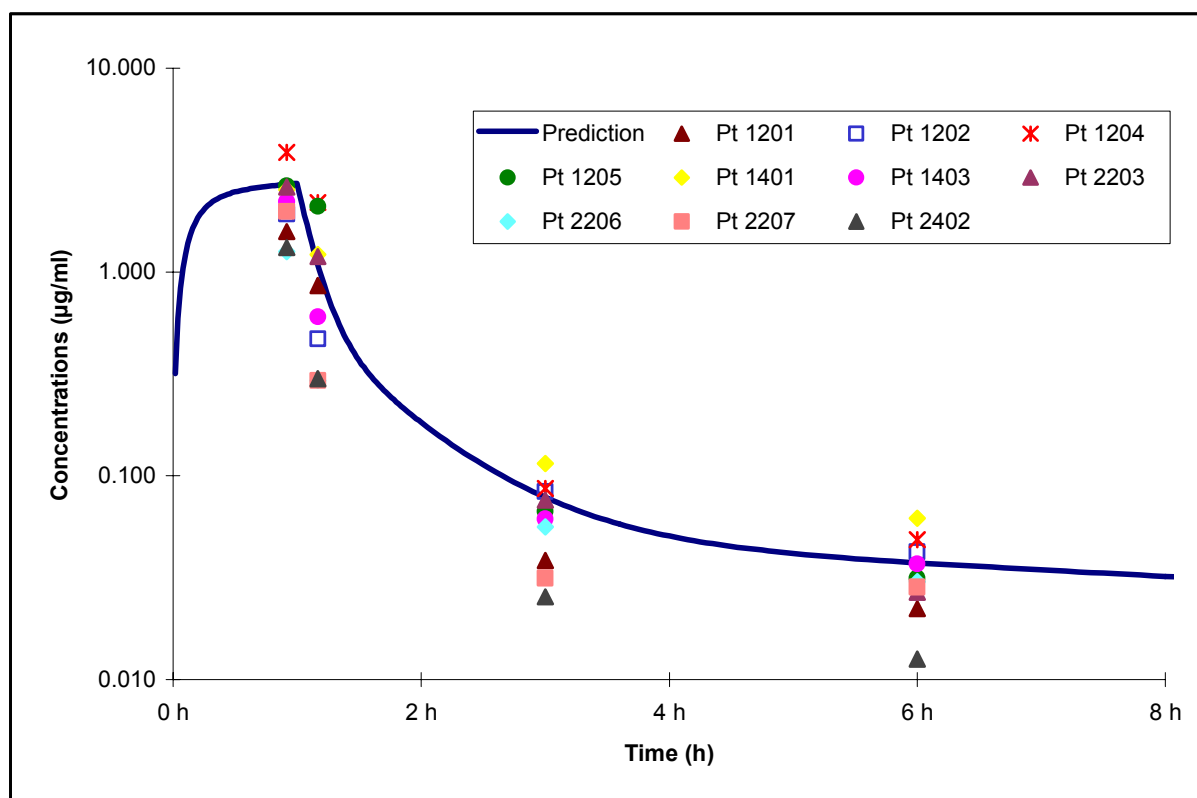
Free and bound aflibercept concentrations are comparable to those observed in a monotherapy study, suggesting that docetaxel has no influence on the pharmacokinetics of aflibercept. The

kinetics of free aflibercept is dose-independent (clearance does not change with dose). Bound aflibercept concentrations estimated at the end of cycle 1 are approximately two times lower than that measured all along the following cycles.

The ratio of free/ bound aflibercept was maintained above 1 at the end of cycle 1 administration, for all the patients from dose level 6mg/kg.

Furthermore, the preliminary results of docetaxel are compared to a pharmacokinetic profile given by the population PK and illustrated hereunder in [Figure 2](#).

Figure 2 - Individual plasma concentrations of docetaxel in comparison with the prediction given by the population pharmacokinetics



Preliminary data indicate that aflibercept has likely no impact on the pharmacokinetics of docetaxel.

These preliminary data indicate the absence of interaction between docetaxel and aflibercept.

4.4.3.3 Conclusion

Aflibercept and docetaxel in combination were well tolerated with acute and reversible adverse reactions linked to VEGF blockade (mainly HBP), no cumulative or late toxicity were observed. In addition, no evidence of exacerbation of background chemo-related toxicities was observed. The safety profile appears to be quite similar across the 4 dose levels explored even at the highest

dose tested. Antitumor activity was observed at each dose level. Around one third of patients had a HBP requiring treatment from the dose of 5 mg/kg and above. A trend for more severe episodes of HBP is observed at the two highest dose level explored (7 and 9 mg/kg). The criteria for recommended-dose selection took into account the combination of DLTs/overall safety profile and pharmacokinetics together. The aflibercept dose of 6 mg/kg in combination with docetaxel 75 mg/m², given every 3 weeks has been selected, and the expansion cohort is ongoing.

4.5 RATIONALE FOR THE STUDY

Aflibercept has shown evidence of antitumor activity as a single agent in patients with advanced lung cancer. In addition, aflibercept is anticipated to be more active than other anti-VEGF agents because of its particular high binding affinity to VEGF.

Docetaxel single agent, 75 mg/m² given every 3 weeks, is an approved regimen for the treatment of patients with locally advanced or metastatic non-small cell lung cancer after failure of prior platinum-based chemotherapy, and therefore represents a valid reference treatment.

There are consistent preclinical and clinical evidences to support the conduct of this study assessing the potential benefit of combining aflibercept to docetaxel in patients with NSCLC having failed first-line platinum-based chemotherapy.

The doses for the combination of aflibercept (6 mg/kg) and docetaxel (75 mg/m²) given every 3 weeks were selected based on the results of the previous human experience data reported above.

5 STUDY OBJECTIVES

5.1 PRIMARY

To demonstrate overall survival (OS) improvement for aflibercept + docetaxel compared to docetaxel + placebo as second line treatment for patients with locally advanced or metastatic non-small cell lung cancer (NSCLC).

5.2 SECONDARY

The secondary objectives are:

- To compare efficacy of aflibercept to placebo for:
 - Progression Free Survival (PFS),
 - Response Rate (RR) as per RECIST criteria [24] ([Appendix B](#)),
 - Health Related Quality of Life (HRQL) assessed by the Lung Cancer Symptom Scale (LCSS) questionnaire [28] [Appendix G](#). Questionnaires will be self-administered by the patients. Visual analogue scales with 100-mm lines are used to measure the intensity of patient responses,
- To assess the overall safety of the two treatment arms,
- To assess the pharmacokinetics of intravenous (IV) aflibercept,
- To determine immunogenicity of IV aflibercept (anti-aflibercept antibody detection) in all patients.

6 STUDY DESIGN

This is a prospective, multicenter, multinational, randomized (1:1), double-blind, parallel-group study comparing the efficacy in term of overall survival (OS) of aflibercept (Arm A) versus placebo (Arm B) administered on top of the standard docetaxel regimen as second-line treatment for patients with NSCLC.

6.1 DESCRIPTION OF THE PROTOCOL

Each patient will be treated every 3 weeks, in the absence of definitive treatment discontinuation criteria outlined in Section 11.1.

An executive Steering Committee will be responsible for supervising the progress of the trial. This committee will include the Study Chairman, designated main Investigators and the Sponsor's representatives.

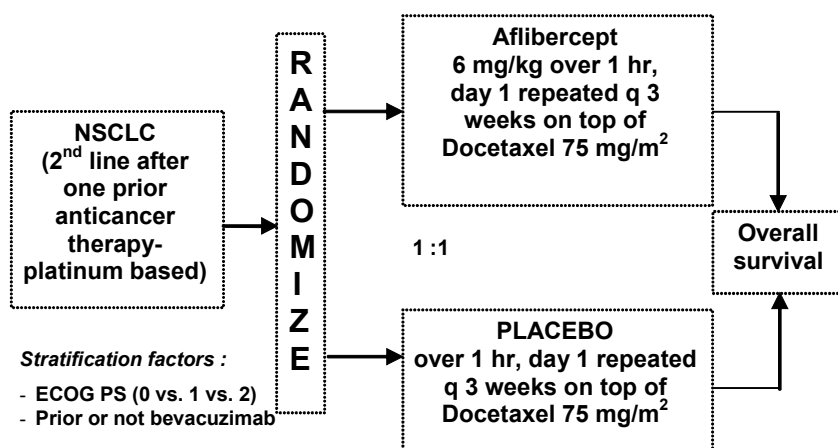
An independent Data Monitoring Committee (DMC) will periodically assess the patients' safety data, risk / benefit to the patients and will advise the Steering Committee and Sponsor regarding the continued safety of treating ongoing and future study patients. **In addition the DMC will review the efficacy data from the planned survival interim analysis (IA).**

Allocation to the two treatment arms will be done centrally via an Interactive Voice Response System (IVRS), using a permuted-block randomization according to the following factors:

- Eastern Cooperative Oncology Group (ECOG) performance status (0 vs.1 vs. 2),
- Prior or not bevacizumab.

The study design is summarized in Figure 3:

Figure 3 - Graphical Study Design



Flowchart - please refer to study flowcharts in Section 1.1 and Section 1.2.

6.2 DURATION OF STUDY PARTICIPATION

Patients will be considered on study upon signing the informed consent. As shown in the graphical study design and study flowchart (Sections 1.1 and 1.2), the study consists of a 21-day screening phase prior to randomization, start of study treatment administration randomly assigned within 3 days of randomization, 21-day study treatment cycles, safety evaluation on Day 30 and a follow-up phase (every two months after last study treatment administration).

During the screening period, all baseline procedures (see Sections 1.1 and 1.2) will have to be performed within defined timelines, including review of eligibility criteria (see section 7.2 and section 7.3).

During the treatment period, patients will be treated until one of the following events occurs: disease progression, unacceptable toxicity, or patient refusal of further study treatment. The study treatment, either aflibercept or placebo (combined with docetaxel), should be administered every 3 weeks and the patients will be followed for safety for a minimum of 30 days following the last administration of the study drug (either aflibercept/placebo or docetaxel). Cycle lengths may be extended in case of unresolved toxicity (see section 8.4.3). All patients' study drug-related AEs and all SAEs should be followed until resolution/stabilization. Possible study drug-related SAEs brought to the attention of the investigator at any time after cessation of study medication should be reported to the monitoring team.

During the post-treatment follow-up period, all patients will be followed for disease status (including radiological assessments), every 6 weeks, until disease progression, as defined in Section 9.1.2.1, is documented (for a maximum of 12 months) or until the study cut-off date, whichever comes first.

After disease progression each patient will be followed every 2 months for survival status, and collection of data regarding further anticancer therapy, until death or the study cut-off date, whichever comes first.

The maximum duration of the study and the final data cut-off date for survival will be the first date when it is determined that approximately 683 deaths have occurred.

6.3 INTERIM ANALYSIS

One interim analysis (IA) of overall survival is planned to stop for futility or overwhelming efficacy when approximately 344 OS events (50% information fraction) have occurred. Statistical operating characteristics of this analysis e.g. stopping boundaries and α - and β -spending functions to maintain the type I and type II error rates are described in the statistical considerations (see section 13.6).

The interim analysis will be reviewed by the DMC.

6.4 STUDY COMMITTEES

An **Executive Steering Committee** will include the Study Chairman, the designated main Investigators and Sponsor representatives and be responsible for:

- Supervising the progress of the trial towards its overall objectives,
- Reviewing at regular intervals relevant information that may affect the study conduct,
- Discussing the implementation of the recommendations of the independent Data Monitoring Committee.

The Executive Steering Committee will have no access to unblinded data **before completion of study, except if released by the DMC at the time of IA for overall survival (see Section 9.4).**

An independent **Data Monitoring Committee (DMC)** consisting of at least 3 external independent members not associated with the conduct of the study or other study committees will meet regularly to:

- Review the progress of the trial,
- Review the unblinded safety data,
- Advise the Sponsor and the Executive Steering Committee on potential modifications or communications that may be necessary to ensure the patient safety or protect the scientific integrity of the trial.

The first safety review of the trial by the DMC will be the earlier of either 6 months after the first patient has been enrolled, or the time 100 patients (50 patients /arm) treated with at least 3 cycles. The DMC meeting will be set up semiannually. Ad-hoc DMC meetings may also be held if a significant safety issue or issue deemed important for discussion arise on this or any other studies of aflibercept. After each meeting, the DMC will advise the Executive Steering Committee and the Sponsor's representatives regarding the patients' safety, as well as the course of action regarding the conduct of the trial.

The DMC procedures will be detailed in the DMC charter and approved by the DMC members.

7 SELECTION OF PATIENTS

7.1 NUMBER OF PATIENTS PLANNED

Approximately 900 patients with NSCLC, 450 patients per arm, are expected to be enrolled and treated in this study, (please refer to statistical Section 13.2). It is planned to recruit this sample from approximately 120 sites within 18 months. No patient may be randomized into the study more than once.

7.2 INCLUSION CRITERIA

Patients meeting the following criteria will be considered for enrollment into the study :

1. Histological/cytological proven locally advanced or metastatic non-small-cell lung cancer.
2. Disease progression during or after one, and only one, prior anticancer therapy which is platinum-based (chemotherapy or targeted therapy) for advanced or metastatic disease. Disease progression within 6 months of adjuvant platinum-based chemotherapy is accepted.

7.3 EXCLUSION CRITERIA

Patients presenting with any of the following will not be included in the study :

Related to methodology:

1. Squamous histology/cytology.
2. Less than 28 days elapsed from prior treatment with radiotherapy, surgery or chemotherapy to the time of randomization. Less than 42 days elapsed from prior major surgery (lung resection) to the time of randomization.
3. Prior isotope therapy, whole pelvic radiotherapy, or radiotherapy to > 25% of bone marrow ([Appendix F](#)).
4. Adverse events (excluding alopecia and those listed in the specific exclusion criteria) from any prior anticancer therapy of grade >1 (National Cancer Institute Common Terminology Criteria [NCI CTCAE] v.3.0) at the time of randomization ([Appendix E](#)).
5. Age < 18 years.
6. Eastern Cooperative Oncology Group (ECOG) performance status (PS) > 2 ([Appendix D](#)).
7. History of brain metastases, uncontrolled spinal cord compression, or carcinomatous meningitis, or new evidence of brain or leptomeningeal disease.
8. History of another neoplasm. Adequately treated basal cell or squamous cell skin cancer, or in situ cervical cancer, or any other cancer from which the patient has been disease-free for > 5 years are allowed.

9. Participation in another clinical trial and any concurrent treatment with any investigational drug within 30 days prior to study randomization.
10. Any of the following events within the 3 months prior to randomization: treatment resistant peptic ulcer disease, erosive oesophagitis or gastritis, grade 3 or 4 gastrointestinal bleeding/hemorrhage, infectious or inflammatory bowel disease, diverticulitis, pulmonary embolism, or other uncontrolled thromboembolic event.
11. Any of the following events within the 6 months prior to randomization: myocardial infarction, severe/unstable angina pectoris, coronary/peripheral artery bypass graft surgery, NYHA class III or IV congestive heart failure ([Appendix H](#)), stroke or transient ischemic attack
12. Occurrence of deep vein thrombosis within 4 weeks, prior to randomization.
13. Acquired immunodeficiency syndrome (AIDS-related illnesses) or known Human Immunodeficiency Virus (HIV) disease requiring antiretroviral treatment.
14. Any severe acute or chronic medical condition, which could impair the ability of the patient to participate to the study or interfere with interpretation of study results.
15. Absence of signed and dated Institutional Review Board (IRB)-approved patient informed consent form prior to enrollment into the study.
16. Pregnant or breast-feeding woman. Positive serum or urine pregnancy test prior to randomization
17. Patient with reproductive potential (M/F) who do not agree to use accepted and effective method of contraception during the study treatment period and for at least 6 months after the completion of the study treatment. The definition of “effective method of contraception” will be based on the investigator’s judgment.

For female patients enrolled in the United Kingdom (unless surgically sterile, post-menopausal or for another reason have no chance of becoming pregnant) the following forms of contraception are acceptable: oral contraceptives accompanied by the use of a second method of contraception, as it is not known how oral contraceptives interact with all the study medications or Intra Uterine Device (IUD). For male patients their partner should use an effective means of contraception as described above.

Related to aflibercept:

18. History of prior discontinuation of any anti-VEGF agent due to adverse drug reaction.
19. Urine protein:creatinine ratio (UPCR) > 1 on morning spot urinalysis or proteinuria > 500 mg/24h
20. Serum Creatinine > 1.5 x ULN (if creatinine 1.0 - 1.5 x ULN, creatinine clearance calculated according to Cockcroft-Gault formula < 60 mL/min will exclude the patient ([\[29\]](#), see [Appendix A](#)).
21. Uncontrolled hypertension, defined as blood pressure >150/100 mm Hg (grade ≥ 2 according to NCI CTCAE v.3.0), or systolic blood pressure >180 mm Hg if diastolic blood pressure <90

mm Hg, on at least 2 repeated determinations on separate days, within 3 months prior to study randomization.

22. Patients on anticoagulant therapy with unstable dose of warfarin and/or having an out-of-therapeutic range INR (> 3) within 4 weeks prior to randomization.
23. Evidence of clinically significant bleeding diathesis including hemoptysis or underlying coagulopathy (e.g. INR > 1.5 without vitamin K antagonist therapy), non-healing wound.

Related to docetaxel:

24. Prior docetaxel treatment.
25. Patients refractory to first line paclitaxel-based therapy (patients are considered refractory if their best response on treatment was Progressive Disease).
26. History of hypersensitivity to docetaxel, or polysorbate 80.
27. Inadequate organ and bone marrow function as evidenced by:
 - Hemoglobin < 10.0 g/dL
 - Absolute neutrophil count $< 1.5 \times 10^9/L$
 - Platelet count $< 100 \times 10^9/L$
 - AST/SGOT and/or ALT/SGPT $> 2.5 \times ULN$
 - Alkaline phosphatase (AP) $> 5 \times ULN$
 - AST/SGOT and/or ALT/SGPT $> 1.5 \times ULN$ concomitant with AP $> 2.5 \times ULN$
 - Total bilirubin $> 1.0 \times ULN$
28. Contraindications to the use of corticosteroid treatment.
29. Symptomatic peripheral neuropathy grade > 2 (National Cancer Institute Common Terminology Criteria [NCI CTCAE] v.3.0).

8 TREATMENT

8.1 DETAILS OF TREATMENTS

8.1.1 Names, formulations and manufacturers

Study treatment details are summarized in [Table 6](#) below.

Table 6 - Details of treatments

Drug code	AVE0005 (VEGF Trap)	Placebo for AVE0005 (VEGF Trap)	RP56976
INN	Aflibercept	Not applicable	Docetaxel
Trade name	-	Not applicable	Taxotere
Formulation	Aflibercept in 5 mM phosphate, 5 mM citrate, 100 mM sodium chloride, 20% (w/v) sucrose, and 0.1% (w/v) polysorbate 20, pH 6.0, supplied in sealed, sterile, single-use 5-mL vials containing 4.4-mL in order to withdraw 4 mL aflibercept at a concentration of 25 mg/mL.	Sterile aqueous buffered vehicle pH 6.0, containing 5 mM phosphate, 5 mM citrate, 100 mM sodium chloride, 0.1% (w/v) polysorbate 20 and 20% (w/v) sucrose, supplied in sealed, sterile, single-use 5-mL vials containing 4.4-mL in order to withdraw 4 mL placebo at a concentration of 25 mg/mL.	Marketed formulation (refer to the local labeling)
Storage conditions	Aflibercept must be refrigerated at 2–8°C (36–46°F) in a locked area with restricted access and handled in accordance with the manufacturer's instructions.	Placebo must be refrigerated at 2–8°C (36–46°F) in a locked area with restricted access and handled in accordance with the manufacturer's instructions.	Store between 2–25°C (36–77°F). Retain in the original package to protect from bright light. Freezing does not adversely affect the product.

8.2 INVESTIGATIONAL PRODUCT

8.2.1 Description of aflibercept

Aflibercept is formulated as a sterile liquid to a final concentration of 25 mg/mL. Aflibercept in 5 mM phosphate, 5 mM citrate, 100 mM sodium chloride, 20% sucrose, and 0.1% polysorbate 20, pH 6.0 will be supplied in sealed, sterile, single-use 5-mL vials containing 4.4-mL in order to withdraw 4 mL.

8.2.2 Description of placebo for aflibercept

Placebo for aflibercept is formulated in bulk aqueous buffered solution, pH 6.0, containing 5 mM phosphate, 5 mM citrate, 100 mM sodium chloride, 0.1% (w/v) polysorbate 20, and 20% (w/v)

sucrose, supplied in sealed, sterile, single-use 5-mL vials containing 4.4-mL placebo in order to withdraw 4mL.

8.2.3 Preparation reconstitution and administration for aflibercept and placebo

Sealed, sterile, single-use 5-mL vials containing 4.4-mL with a withdrawable content of 4.0-mL aflibercept/placebo at a concentration of 25 mg/mL will be supplied by the Sponsor.

Aflibercept/placebo must be diluted in 0.9% NaCl (normal saline) or 5% dextrose prior to IV administration.

The dilution must be carried out under aseptic conditions.

Diluted aflibercept/placebo solution at 0.6 to 8.0 mg/ml can be stored up to 24 hours under refrigerated conditions (2° to 8°C) or for up to 8 hours at ambient temperature (approximately 25°C) in polypropylene syringe or in infusion bags made of the following materials:

- PVC containing DEHP
- Polyolefin (PVC free DEHP free)

Diluted solutions of aflibercept/placebo can be administered using infusion tubing made of the following materials:

- PVC containing DEHP
- DEHP-free PVC containing TOTM
- Polypropylene
- Polyethylene lined PVC

The infusion sets must contain a 0.2 µm polyethersulfone filter.

The research pharmacist or designee at the study site will prepare all aflibercept/placebo infusion solutions according to the instructions contained in protocol supporting documents. Multiple vials may be required in the preparation of each dose to obtain a dosing solution at a concentration of 0.6 to 8.0 mg/mL.

The volume of drug to be administered to each patient, and hence the rate of infusion, will be based on each patient's weight and intended dose, patient's weight needs to be reevaluated before each administration. The pharmacist or designee will prepare the dosing solutions according to one of the following 2 methods :

- Calculate the number of vials needed according to body weight and intended dose.

Then, either method 1:

- Dilute the entire volume of each vial in the infusion bag of 0.9% NaCl or 5% dextrose,
- Retrieve the excess of the diluted solution in order to obtain the exact amount of aflibercept/placebo to be administered to the patient.

Or method 2:

- Withdraw the necessary volume of aflibercept/placebo from each vial,
- Dilute directly into the infusion bag of 0.9% NaCl or 5% dextrose.

Infusion can be conducted by gravity or, with an IV infusion pump, or with a syringe pump using administration sets made of above materials.

The infusion should not exceed 2 hours at ambient temperature (approximately 25°C).

8.2.4 Precautions

All drug packages are to be inspected upon receipt at the study site and the individual vials inspected prior to being drawn up. If a vial is chipped or any particulate matter or cloudiness is detected, the vial is not to be used. Cloudy or damaged vials are to be reported to the Sponsor and returned to the refrigerator until instructions have been given. As aflibercept is a protein, the study vials are not to be shaken.

Given the investigational nature of the product and to provide the patients with the maximum level of safety in case of an unexpected event, the following requirements must be fulfilled before any administration of the investigational product can start and for a minimum of 1 hour following the completion of the infusion:

1. Immediate access to appropriate resuscitative equipment
2. Appropriately qualified and trained personnel must be on site (see also section 15.1 for delegation of investigator duties).

Aflibercept should not be administered less than 48 hours following minor surgical procedures (e.g., fine needle biopsy/aspiration, placement of a central venous access device, or removal/biopsy of a skin lesion), or until evidence of wound healing (e.g., scab formation) is observed, whichever is longer. A peripheral venous catheter may be inserted prior to and removed immediately after individual study drug doses, for the purpose of study drug administration.

Infusion and hypersensitivity reactions may occur during or shortly after intravenous administration of protein therapeutics. If grade 2 infusion or hypersensitivity reactions occur in a given patient, institutional treatment guidelines for similar therapeutic agents or guidelines provided in (see Table 9) should be followed, aflibercept/placebo should be resumed after patient recovery and samples for anti-aflibercept antibodies detection should be collected within 2 weeks (see Appendix C). If antibodies are detected, new samplings have to be collected every 2-3 months, up to 6 months from last dose. If grade ≥ 3 infusion or hypersensitivity reactions occur in a given patient, institutional treatment guidelines for similar therapeutic agents or guidelines

provided in (see [Table 9](#)) should be followed, aflibercept/placebo should be definitively discontinued and samples for anti-aflibercept antibodies detection should be collected within 2 weeks (see [Appendix C](#)). If antibodies are detected, new samplings have to be collected every 2-3 months, up to 6 months from last dose.

8.3 ASSOCIATED PRODUCT: DOCETAXEL

Marketed formulation of docetaxel will be used.

Refer to the docetaxel package insert or summary of product characteristics for details on description, preparation, administration, and precautions for use.

8.4 DOSAGE SCHEDULE

Patients will be randomly assigned to receive either **arm A**: Aflibercept or **arm B**: Placebo. Within 3 days after randomization, patients will receive either aflibercept or placebo, depending on arm assigned.

Immediately after, patients will receive docetaxel.

This treatment schedule will be repeated every 3 weeks.

8.4.1 Aflibercept/placebo

Aflibercept: 6 mg/kg will be administered IV over 1 hour once on Day 1, every 3 weeks, prepared and administered as described in Section [8.2.3](#).

Or

Placebo: 6 mg/kg will be administered IV over 1 hour once on Day 1 every 3 weeks, prepared and administered as described in Section [8.2.3](#).

8.4.2 Docetaxel regimen

8.4.2.1 Treatment

Immediately after aflibercept/placebo administration, all the patients will receive docetaxel: 75 mg/m² in 250 mL dextrose 5% or NaCl 0.9% IV over 1 hour, once on Day 1 every 3 weeks.

In case of BSA > 2.2 m², the actual dose of docetaxel should be adjusted to a maximum BSA of 2.2 m² for safety reasons.

BSA will be calculated at the start of each treatment cycle from body weight in kg, recorded prior to each treatment cycle, and height in cm, recorded at baseline. The preferred Dubois and Dubois equation is below:

$$\boxed{\text{BSA in units of m}^2 = \text{wgt. in kg}^{0.425} \times \text{hgt. in cm}^{0.725} \times 0.007184}$$

8.4.2.2 Premedication

Corticosteroid pre-postmedication for docetaxel: oral dexamethasone 8 mg (or equivalent) for a total of 6 doses will be administered, evening before Day 1, on Day 1 (early morning, 1 hour before docetaxel treatment, and evening) and on Day 2 (morning and evening). Refer to the package insert of docetaxel for details.

Antiemetic: Appropriate prophylactic antiemetic therapy is left to current hospital practices.

Granulocyte-colony stimulating factor (G-CSF): The use G-CSF is left to the current hospital practices [27].

8.4.3 Schedule modification

For both arms, it is a 3 weekly schedule. Plus or minus 2-day time windows for each treatment day are permitted, as are treatment delays of up to 2 weeks in case of unresolved toxicity. Doses may be modified or infusion delayed for toxicity as described in Section 8.5. New cycles of therapy may not begin until any study drug-related toxicities have been adequately resolved. Study treatment will continue until a definitive treatment discontinuation criterion is met (see section 11.1).

Details of the exact dose, and time of administration of medication (day/month/year, h:min) and the labelling on the investigational product containers (including batch and/or lot numbers) will be documented in the case report form (CRF). In addition, complete Drug Accountability Records must be maintained at each site.

8.5 DOSAGE MODIFICATION

Dose adjustment and/or cycle delay are planned in case of toxicity. Dose adjustments will be made according the worst grade of toxicity. Toxicities will be graded according the NCI-CTC AE V. 3.0 scale ([25], Appendix E). Patient will receive the next cycle after recovery of the toxicity.

If a patient experiences several toxicities and there are conflicting recommendations, the most conservative dose adjustment recommended (dose reduction appropriate to the most severe toxicity) should be followed. Once a dose has been decreased, intra-patient re-escalation back to the previous dose level is not permitted.

No more than 1 dose reduction, or 1 dose omission, as well as no more than 2 week delay in infusion of aflibercept/placebo per patient are permitted. Aflibercept/placebo will be discontinued in patients requiring > 1 dose reduction of aflibercept/placebo or > 2 week delay due to aflibercept/placebo toxicity. Omission is planned in certain circumstances described in Table 10.

No more than 2 dose reductions as well as no more than 2 week delay in infusion of docetaxel per patient are permitted. Docetaxel will be discontinued in patients requiring > 2 dose reductions of docetaxel or > 2 week delay due to docetaxel toxicity.

- If docetaxel is permanently discontinued, then aflibercept/placebo will be continued alone until disease progression or unacceptable toxicity or patient refusal of further treatment. The end of study treatment will be the date of the last aflibercept or placebo administration.
- If aflibercept/placebo is permanently discontinued, then docetaxel will be continued alone until disease progression or unacceptable toxicity or patient refusal of further treatment. The end of study treatment will be the date of the last docetaxel administration

In both cases the end of the study is the date of death, cut off or withdrawal of consent, whichever comes first.

8.5.1 Aflibercept/placebo

Aflibercept/placebo doses will be modified according to the dose level described in [Table 7](#) below. Only one dose reduction is permitted.

Table 7 – Aflibercept /placebo Dose reduction level

	Initial dose (mg/Kg)		Dose reduction 1 (mg/Kg)
Aflibercept /placebo	6	→	3

Actions to be taken for aflibercept/placebo according to the type of toxicity are described in [Table 9](#) and in [Table 10](#).

Table 8 – Dose modifications for aflibercept/placebo

Toxicity	Grade	Action to be taken
Hypertension (see below)	Grade ≤ 2	Initiate antihypertensive drug therapy (see recommendation below) and closely monitor BP for further therapy adjustment as needed. No dose modification and no delay.

Toxicity	Grade	Action to be taken
	Grade 3 (requiring more than one drug or more intensive therapy than previously)	Modify antihypertensive drug therapy (see recommendation below). Delay the administration of both docetaxel and aflibercept/placebo, for a maximum of 2 weeks, until recovery to BP $\leq$ 150/100 or to systolic BP < 180 if diastolic BP < 90 for patients with known history of isolated systolic hypertension: - If BP is controlled within 2 weeks delay: - First episode: readminister docetaxel and aflibercept/placebo at the same dose. - Second episode: readminister docetaxel and aflibercept/placebo, with aflibercept/placebo reduced to dose level 1*. - Third episode, discontinue aflibercept/placebo. - If BP is still uncontrolled despite appropriate anti hypertensive treatment and after 2 weeks delay: - Administer docetaxel alone and omit aflibercept/placebo for 1 cycle. The reintroduction of aflibercept/placebo at a dose reduced to dose level 1* will be reconsidered at the time of the administration of the subsequent cycle (in combination with docetaxel), only if BP is controlled at the time of re-administration. - In case of reoccurrence of grade 3 BP despite dose reduction of aflibercept/placebo, or if BP is still uncontrolled despite 1 omission of administration of aflibercept/placebo, the patients will be discontinued from aflibercept /placebo. Docetaxel will be continued if the investigator thinks the patient is benefiting from it.
	Grade 4	Consult Cardiologist, and discontinue aflibercept/placebo.
Arterial thromboembolic events (e.g. : Myocardial infarction, or stroke) documented by the appropriate tests	Grade 3-4	Discontinue aflibercept/placebo
Hemorrhage	Grade 3-4	Discontinue aflibercept/placebo
Hemoptysis	Grade 2	Stop aflibercept/placebo and perform a bronchoscopy : in case of underlying high-risk lesion, discontinue aflibercept/placebo. If not and if hemoptysis resolved within 2 weeks resume aflibercept/placebo, otherwise discontinue aflibercept/placebo.
Reversible Posterior Leukoencephalopathy Syndrome documented by the appropriate tests (see below)	Any Grade	Discontinue study treatments
Venous Thromboembolic Event documented by the appropriate tests	Grade 3 (DVT)	1st episode: Treat DVT with heparins and continue study treatment** 2nd episode despite appropriate anticoagulation: Discontinue aflibercept/placebo
	Grade 4 (PE)	Treat PE and discontinue aflibercept/placebo ***
Gastro-intestinal perforation, fistula formation (see below)	Any Grade	Discontinue study treatments

* Dose reduction levels provided in [Table 7](#)

** Based on investigator's judgement in assessing potential risk of extension and/or embolization.

*** Discussion with the Sponsor if investigator wishes to continue aflibercept/placebo in case of asymptomatic, incidentally discovered PE in a patient with a US evidence of DVT

Hypertension therapy recommendations:

For patients without prior antihypertensive therapy, the initiation of calcium-channel blockers should be considered as a first-intent treatment. Ultimately, antihypertensive treatment must be individualized based on the presence of comorbidity such as diabetes, cardiovascular or renal disease. In addition, oral and/or intravenous sodium intake should be carefully monitored in these patients.

BP will be closely monitored for further adjustment in therapy as needed (in between scheduled treatment visit).

For patients already under anti-hypertensive therapy, efforts should be done to optimize (in between scheduled treatment visit) the existing therapy before adding other agents as required to control the blood pressure.

BP will be closely monitored for further adjustment in therapy as needed (in between scheduled treatment visit).

When hypertension is accompanied by signs or symptoms of end organ damage such as hypertensive retinopathy, kidney function abnormalities (like progressive proteinuria), or any signs or symptoms of cardiovascular or central nervous system morbidity, treatment with aflibercept should be interrupted.

Reversible posterior leuko-encephalopathy or clinical symptoms related to vasogenic edema of the white matter:

Clinical presentations are variable and may include headache, altered mental status, seizure and cortical visual deficit. Hypertension is a risk factor. MRI scans are key to diagnosis and typically demonstrate vasogenic edema (hyperintensity in T2 and FLAIR images and hypodensity in T1 images) predominantly in the white matter of the posterior parietal and occipital lobes; less frequently, the anterior distributions and the gray matter may also be involved. RPLS should be in the differential diagnosis in patients presenting with unexplained mental status change, visual disturbance, seizure, or other CNS findings. RPLS is potentially reversible with early recognition of symptoms and timely correction of the underlying causes, including control of blood pressure and interruption of the offending drug, which are important in order to prevent progression to irreversible tissue damage.

Gastro-intestinal Perforation:

If a patient develops new or increase in severity of abdominal pain, with or without accompanying symptoms such as: vomiting, nausea and constipation, the patient should be evaluated by a physician for possible gastro intestinal perforation. This has been reported with anti-VEGF agents.

Hypersensitivity reaction:

In case of hypersensitivity reaction, institutional treatment guidelines for this type of adverse event, or the guidelines proposed in [Table 9](#) can be applied. Venous blood samples for anti-aflibercept antibody measurements should be collected in any patient developing grade ≥ 2 systemic immunologic adverse event considered at least possibly related to study drug.

Table 9 - Acute infusion reaction management

Symptom Severity	Intervention Recommendation
<u>Mild-Moderate</u> e.g., NCI CTCAE grade ≤ 2 cutaneous reaction, pruritus, flushing, rash, dyspnea, tachycardia, hypotension, anxiety, headache, myalgias, edema, nausea	Stop aflibercept/placebo infusion; Give diphenhydramine 50 mg IV and/or IV dexamethasone 10 mg; For Gr 2: Collect blood sample for detection of anti-aflibercept antibodies within a maximum of 2 weeks, and then every 2-3 months for up to 6 months from last aflibercept/placebo dose. Resume aflibercept/placebo infusion after subject recovery.
<u>Severe</u> e.g., symptomatic bronchospasm, generalized urticaria, systolic BP ≤ 80 mm Hg, angioedema, anaphylaxis	Stop aflibercept/placebo infusion; Give IV diphenhydramine 50 mg and/or IV dexamethasone 10 mg and/or epinephrine as needed; Collect blood sample for detection of anti-aflibercept antibodies within a maximum of 2 weeks, and then every 2-3 months for up to 6 months from last aflibercept/placebo dose. Discontinue aflibercept/placebo.

Proteinuria:

Determination and management of proteinuria:

Prior to administration of aflibercept/placebo, perform an UPCR and dipstick.

Urinary protein creatinine ratio (UPCR) corresponds to the ratio of urinary protein and urinary creatinine concentrations (both expressed in mg/dL). There is a high correlation between morning UPCR and 24-hour proteinuria in patients with normal or reduced renal function, UPCR demonstrated very good to excellent performance for the diagnosis of both abnormal and nephrotic proteinuria at all renal function levels. This ratio provides an accurate quantification of 24-hours urinary protein excretion [26].

UPCR to detect proteinuria, will be done on morning urine spot. If UPCR > 1 , 24-hour urine collection to grade proteinuria will be performed. In addition, in case of proteinuria from renal origin (according to urin protein electrophoresis) with UPCR > 2 , or UPCR > 1 associated with hematuria, then a clinical work up to rule out thrombotic microangiopathy should be performed : it may include LDH, haptoglobin, schistocytes and orosomucoïd (if available) measured in blood. Anti-aflibercept antibodies will be assessed (a sample for aflibercept/placebo pharmacokinetic will be taken in parallel) and a nephrologist advice should be considered as detailed in [Table 10](#). Anti-aflibercept antibodies will be assessed every 2-3 months for up to 6 months from last aflibercept/placebo dose.

Proteinuria should always be assessed taking into account the presence or absence of hematuria and the blood pressure status of the patient.

Actions to be taken are described in [Table 10](#).

Table 10 - Proteinuria management

	Cycle n	24h -Proteinuria performed on D1 of cycle n	24h -Proteinuria performed within cycle n	Cycle n+1	24h-Proteinuria prior Cycle n+2	Cycle n+2
If UPCR [0-1]	Afibercept°: same dose Docetaxel: same dose					
If UPCR [1-2] without Hematuria	Administer the cycle with no delay Afibercept°: same dose Docetaxel: same dose then	≤ 3.5g [Gr1 or 2]	If Proteinuria return to ≤ 2g within 2 weeks	Afibercept°: same dose Docetaxel: same dose		
			If Proteinuria return to ≤ 2g between 2-3 weeks	Afibercept°: dose level 1* Docetaxel: same dose		
			If Proteinuria >2g – ≤ 3.5g	Omit Afibercept° Docetaxel : same dose	> 2g	Stop Afibercept° Docetaxel : same dose
					≤ 2g	Afibercept° dose level 1* Docetaxel: same dose
	Perform 24h urine collection to assess the proteinuria rate and urinary protein electrophoresis. 24h proteinuria will be repeated weekly	> 3.5 g** [Gr3]	If Proteinuria ≤ 2g	Afibercept° level 1* Docetaxel : same dose		
			If Proteinuria >2g – ≤ 3.5g	Omit Afibercept° Docetaxel : same dose	> 2g	Stop Afibercept° Docetaxel : same dose
					≤ 2g	Afibercept° dose level 1* Docetaxel : same dose
			If Proteinuria >3.5 g	Discontinue Afibercept° Docetaxel : same dose		

° Afibercept stands for aflibercept/placebo; * Dose reduction levels provided in [Table 7](#); ** collect blood samples for detection of anti-aflibercept antibodies.

Proteinuria management (to be continued)

	24h -Proteinuria value	Cycle n	24h -Proteinuria at the time of administration of cycle n+1	Cycle n+1	24h-Proteinuria prior Cycle n+2	Cycle n+2
If UPCR > 2 or If UPCR [1-2] with Hematuria then Delay cycle n for a maximum of 2 weeks and Perform biological work- up** and Perform 24h urine collection to assess the rate of proteinuria and urinary protein electrophoresis. 24h proteinuria will be repeated weekly and Nephrologist consultation	If TMA is ruled out and If Proteinuria return to ≤ 2g within 2 weeks	When proteinuria return to ≤ 2 g administer: Aflibercept°: same dose Docetaxel: same dose	≤ 2g	Aflibercept° level 1* Docetaxel: same dose		
	If TMA is ruled out and If Proteinuria > 2g after 2 weeks delay	Omit Aflibercept° Docetaxel: same dose	> 2g	Stop Aflibercept° Docetaxel: same dose		
	If TMA is diagnosed Or nephrotic syndrome [Gr 4]	Discontinue Aflibercept° Administer docetaxel as appropriate with a delay ≤ 3 weeks upon nephrologist advice				

° Aflibercept stands for aflibercept/placebo; * Dose reduction levels provided in [Table 7](#).

** Perform: urinary protein electrophoresis & 24H urine collection to assess proteinuria rate & collect blood samples for haptoglobin + orosomucoid + schistocytes + LDH + detection of anti-aflibercept antibodies.

8.5.2 Docetaxel

Patients will be monitored closely for toxicity. In addition to optimizing supportive care as described in Section 8.13, docetaxel dosages may be adjusted after the first cycle of therapy according to Table 11. Further dosing is acceptable only if the patient has recovered to grade ≤ 1 , provided that definitive treatment discontinuation criteria described in Section 11.1.1 have not been met.

Table 11 - Docetaxel dose reduction levels

	Initial dose (mg/m ²)		Dose reduction 1		Dose Reduction 2
Docetaxel	75	→	60	→	45

Modifications for docetaxel are based on the last prior date of docetaxel administration. For example, a dose delay of up to 2 weeks refers to a dose delay of up to 2 weeks since the last prior date of docetaxel administration.

In case of fever during any cycle, perform blood count and prescribe antibiotherapy and apply dose modification.

Both Table 12 and Table 13 describe recommended chemotherapy dose modifications on Day 1 of a new cycle based on the worst toxicity encountered during the previous cycle:

Table 12 - Chemotherapy dose modifications for hematologic toxicity*

Hematologic Toxicity	Grade 3	Grade 4
Neutropenia If duration ≥ 7 days, or if complicated by $T \geq 38.5^{\circ}\text{C}$, or $T \geq 38.1^{\circ}\text{C} \times 3$ during a 24-hour period, or infection	Delay** infusion until ANC $\geq 1.5 \times 10^9/\text{L}$: 1 st episode: administer prophylactic treatment with GCSF in subsequent cycles 2 nd episode: Reduce docetaxel by 1 dose level 3 rd episode: Reduce docetaxel by a second dose level 4 th episode: Discontinue docetaxel	
Thrombocytopenia	Delay** infusion until platelets $\geq 75 \times 10^9/\text{L}$. No dose reduction is required	Delay** infusion until platelets $\geq 75 \times 10^9/\text{L}$ and reduce docetaxel by 1 dose level. Discontinue docetaxel in case of recurrence.

*Dose reduction levels provided in Table 11

**Delay infusion by maximum of 2 weeks until recovery to grade ≤ 1 (repeat blood count every 2 days) and apply dose reduction according to worst grade observed

Table 13 - Chemotherapy dose modifications for non-hematologic toxicity*

Non-hematologic toxicity§	Grade 2	Grade 3	Grade 4
Diarrhea	No dose reduction required	1 st episode: Reduce docetaxel by 1 dose level 2 nd episode: Reduce docetaxel by a second dose level 3 rd episode: Discontinue docetaxel	
Stomatitis	No dose reduction required	1 st episode: Reduce docetaxel by 1 dose level 2 nd episode: Reduce docetaxel by a second dose level 3 rd episode: Discontinue docetaxel	
Cutaneous Reactions	No dose reduction required	1 st episode: Reduce docetaxel by 1 dose level 2 nd episode: Discontinue docetaxel	Discontinue docetaxel
Peripheral Neuropathy	Reduce docetaxel by 1 dose level	Discontinue docetaxel	
Ototoxicity	No dose reduction required	Discontinue docetaxel	
Transaminase and Alkaline Phosphatase Elevation	Reduce docetaxel by 1 dose level	Discontinue docetaxel	

§ Delay infusion by a maximum of 2 weeks until recovery to grade ≤ 1 and apply dose reduction according to worst grade observed. Otherwise docetaxel will be permanently discontinued

* Dose reduction levels provided in [Table 11](#)

No dose reduction is allowed for abnormal bilirubin levels (> 1.0 x ULN) during the study, but the next cycle will be delayed by a maximum of 2 weeks until bilirubin levels ≤ 1.0 x ULN.

Hypersensitivity reaction to docetaxel.

Refer to [Table 14](#) or the institutional guidelines to manage this kind of toxicity.

Table 14 - Treatment recommendation for hypersensitivity reactions to docetaxel

Severity and symptoms	Treatment recommendations
Mild symptoms: localized cutaneous reactions, e.g., pruritus, flushing, rash	- Decrease the rate of infusion until recovery of symptoms, stay at bedside. - Then, complete docetaxel infusion at the initial planned rate. At subsequent cycles use the same corticosteroid premedication as outlined in package insert.
Moderate symptoms: any symptom not listed above (mild symptoms) or below (severe symptoms), such as generalized pruritus, flushing, rash, dyspnea, hypotension with systolic BP >80 mmHg	- Stop docetaxel infusion. - Give IV antihistamine and IV. corticosteroids^a. - Resume docetaxel infusion after recovery of symptoms. At subsequent cycles, antihistamines^a and steroids^a are to be given i.v., 1 h before infusion, in addition to the corticosteroid premedication planned in package insert

Severity and symptoms	Treatment recommendations
Severe symptoms: such as bronchospasm, generalized urticaria, hypotension with systolic BP $\leq$ 80 mmHg, angioedema	- Stop docetaxel infusion. - Give IV antihistamine and steroids^a. - Add epinephrine^b or bronchodilators and/or IV fluids macro-molecules if indicated. - Once all signs and/or symptoms of HSR disappear, docetaxel may be re-infused within 24 h from the interruption, if medically appropriate, and whenever possible. Premedication regimen as described in package insert is only recommended when docetaxel is re-infused more than 3 h after the interruption. At the subsequent cycles, dexamethasone is to be given at 20 mg orally 24, 18, 13, 7, and 1 h before docetaxel infusion. Additionally diphenhydramine (or equivalent) is to be given at 50 mg IV 1 h before docetaxel infusion. If a severe reaction recurs, discontinue docetaxel.

^a Antihistamines: dexchlorpheniramine intravenous. 5-10 mg, or clemastine intravenous 2 mg, or diphenhydramine intravenous 25-50 mg, or promethazine intramuscular 50-100 mg; Corticosteroids: dexamethasone or equivalent, intravenous 5-10 mg of dexamethasone.

^b Epinephrine: administered at a 1:1000 dilution (0.01 mL per kg with a maximum dose of 0.5 mL subcutaneously repeated every 20 min as necessary).

mmHg = millimeters of mercury; BP = Blood pressure; IV = Intravenous; h = Hour; i.m. = Intramuscular; HSR = Hypersensitivity reaction

8.5.3 Other toxic effects

Any other dose modifications in study treatment that are not described above may be performed at the discretion of the investigator, provided that criteria for patient withdrawal from study treatment described in Section 11.1.1 have not been met.

Study treatment should be held for a maximum of two weeks from the planned date of reinfusion until resolution to $\leq$ grade 1, and then reinstituted, if medically appropriate. A modification of subsequent doses will be considered. These patients will be withdrawn from study treatment if >2 dose reductions for docetaxel or > 1 dose reduction, or > 1 dose omission, for aflibercept/placebo are needed.

8.6 DESCRIPTION OF BLINDING METHODS

Aflibercept and placebo are indistinguishable as described in Section 8.1.1.

Please also refer to Sections 8.10. and 9.4.

8.7 METHOD OF ASSIGNING PATIENTS TO TREATMENT GROUP

Treatment assignment will be performed centrally via the IVRS using a permuted-block randomization stratified according to baseline ECOG performance status (0 vs.1 vs. 2) and prior (yes vs. no) bevacizumab. All eligible patients will be randomly assigned to either the control arm or the experimental arm in a 1:1 ratio.

Once the informed consent signed and after each patient has completed the necessary screening visit procedures, the corresponding baseline CRFs have been completed and the patient is deemed eligible for study entry by the investigator or designee, the study site will contact the IVRS. The site will need to enter the following information regarding the clinical site and study patient:

- Date of informed consent signature
- Patient's date of birth and gender,
- Current ECOG performance status,
- Prior bevacizumab,
- Intended dose of aflibercept/placeb in mg/kg and patient's weight in kg.

The information above will be used to identify the study treatment arm assigned for the patient according to the predefined randomization schedule in blinded fashion by assigning the patient a randomized treatment number corresponding to specific kit number(s) available at the study site containing either aflibercept or placebo.

Details of the IVRS procedure will be provided in the IVRS Site Manual.

Study treatment should begin within 3 days after randomization.

The investigational product will be administered only to patients included in this study following the procedures set out in this clinical trial protocol. Patients withdrawn from the study retain their patient number, and new patients must always be allotted a new patient number.

Before each dose of investigational product during the treatment period, the IVRS will need to be accessed again in order to receive another kit number(s) containing the same investigational product as the one assigned at randomization.

8.8 PACKAGING AND LABELING

Please also refer to Section [9.4](#) .

This is a double-blind study.

Aflibercept or placebo vials will be supplied in identical boxes corresponding to patient kits. Each kit will be labeled with a unique kit number.

Before each cycle, the IVRS will be called and patients will be assigned depending on his/her weight and intended dose (mg/kg) to be administered, 1 or 2 kit numbers corresponding to 1 or 2 boxes available on site.

8.8.1 Aflibercept or placebo

Aflibercept or placebo (as described in Section 8.2) will be packaged by the Sponsor or an approved contractor, in sealed, sterile, single-use vials. The vials containing aflibercept or placebo will be labeled in accordance with the local regulatory specifications.

Batch reference (packaging number, batch number, or lot number) and quantity of vials dispensed to each patient must be recorded in the CRF/drug accountability form.

8.8.2 Docetaxel (Taxotere®)

Commercially available docetaxel will be used.

Batch reference (packaging number, batch number, or lot number) and quantity of docetaxel dispensed to each patient must be recorded in the CRF/drug accountability form.

8.9 STORAGE CONDITIONS

Aflibercept or placebo must be refrigerated at 2 - 8°C (36 - 46°F) in a locked area with restricted access and handled in accordance with the manufacturer's instructions.

For docetaxel refer to the respective package insert or summary of product characteristics.

8.10 ACCESS TO THE RANDOMIZATION CODE DURING THE STUDY

Please refer to Section 9.4: measures to protect the blinding of the trial.

The investigational products will be administered only to patients included in this study following the procedures set out in this clinical trial protocol. Patients withdrawn from the study retain their subject number, and new patients must always be allotted a new subject number.

In case of an Adverse Event, the code can be broken by the Investigator only in exceptional circumstances when knowledge of the Investigational Product is essential for treating the patient. If possible, a contact should be initiated with the Monitoring Team before breaking the code.

The IVRS center should be called if code breaking is necessary. Only if this fails, code-breaking material can be opened.

For each patient, code-breaking material is supplied, containing the name of the treatment. Each treatment box will be labeled with a 3-panel label. The third panel, which is masked by a scratch off laminate, contains treatment (aflibercept or placebo) information. It will be kept in a safe place on site throughout the Clinical Trial. The Sponsor will retrieve all code-breaking material (opened or sealed) on completion of the Clinical Trial.

If the blind is broken, the Investigator will document the date, time of day, and reason for code breaking, and the patient will discontinue permanently the study treatments (see section 11.1.1).

8.11 RESPONSIBILITIES

The Investigator, the Hospital Pharmacist, or other personnel allowed to store and dispense Investigational Product will be responsible for ensuring that the aflibercept/placebo used in the clinical trial is securely maintained as specified by the Sponsor and in accordance with the applicable regulatory requirements.

All aflibercept/placebo shall be dispensed in accordance with the Investigator's prescription and it is the Investigator's responsibility to ensure that an accurate record of aflibercept/placebo issued and returned is maintained.

Any quality issue noticed with the receipt or use of an aflibercept/placebo (deficient IP in condition, appearance, pertaining documentation, labeling, expiry date, etc.) should be promptly notified to the Sponsor, who will initiate a complaint procedure.

Under no circumstances will the Investigator supply aflibercept/placebo to a third party, allow aflibercept/placebo to be used other than as directed by this Clinical Trial Protocol, or dispose of aflibercept/placebo in any other manner.

8.12 RETRIEVAL AND/OR DESTRUCTION OF TREATMENTS

All used treatment vials (aflibercept/placebo and/ or docetaxel) will be destroyed by the study site after an accurate accountability has been performed and signed by the investigator.

All partially used or unused vials of aflibercept/placebo treatments will be retrieved by the Sponsor. A detailed treatment log of the returned Investigational Product will be established with the Investigator (or the pharmacist) and countersigned by the Investigator and the Monitoring Team.

The Investigator will not destroy the unused aflibercept/placebo unless the Sponsor provides written authorization.

A potential defect in the quality of aflibercept/placebo may be subject to initiation by the Sponsor of a recall procedure. In this case, the Investigator will be responsible for promptly addressing any request made by the Sponsor, in order to recall aflibercept/placebo and eliminate potential hazards.

8.13 CONCOMITANT TREATMENT

All treatments being taken by the patient on entry to the study or at any time during the study in addition to the study treatments are regarded as concomitant treatments and must be documented on the appropriate pages of the CRF (as defined in the flow chart section [1.2](#)).

All patient treatments 30 days prior to randomization onto the study (i.e. prior medications), at any time during the study (concomitant medication), and up to 30 days after the end of the study treatment (i.e. post medication) will be considered as concomitant treatments. Any treatment

administered following the withdrawal from the study treatment that are used to treat new or unresolved events related to the study treatment, are as well to be considered as concomitant treatment.

Concomitant medications should be kept to a minimum during the study. However, if these are considered necessary for the patient's welfare and are unlikely to interfere with the investigational products, they may be given at the discretion of the investigator and recorded in the case report form.

8.13.1 Treatments permitted during study treatment

- All supportive measures (including blood transfusions) consistent with optimal patient care will be given throughout the study and should be documented in the CRF.
- Therapeutic or secondary prophylactic use of hematopoietic growth factors may be given at the investigator's discretion and should follow American Society of Clinical Oncology guidelines for their use [27].
- Antihypertensive medications are permitted as described in Section 8.5.1.
- Antiemetic and corticosteroid medication before docetaxel treatment must be administered. The addition of a benzodiazepine, e.g., lorazepam IV or orally, may also be considered if clinically indicated.
- Medications for chronic pain management, including narcotic analgesics, are permitted as clinically indicated.
- Heparins medications are permitted as clinically indicated.

8.13.2 Treatments to be administered with caution

- Vitamin K antagonists.

8.13.3 Treatments not permitted during this study treatment

- Concomitant anticancer therapies, investigational therapies and investigational devices.
- Concomitant radiotherapy.

Patients receiving any of the above prohibited concomitant medications or treatments at the time of screening visit will be ineligible to enter the study until administration of the prohibited agent can be safely discontinued and an appropriate period of time has elapsed as per inclusion/exclusion criteria to permit dissipation of any clinical effects of the drug.

If a patient's clinical status requires administration of a prohibited concomitant medication or treatment, then administration of study drug should be stopped, and the patient will be withdrawn from the study treatment. The change in clinical status mandating the use of the medication in question must be reported as the reason for study drug discontinuation.

8.14 POST-STUDY TREATMENT

Patients will continue to be treated as long as they are benefiting from study treatment and have not met definitive treatment discontinuation criteria as defined in Section 11.1. The data will be collected up to final analysis. The study will be formally closed after all patients have completed the study and the primary analysis has been performed.

Please note that in the absence of documented progressive disease, patients should be followed every 6 weeks until progression is documented or until the study cut-off date, whichever comes first. Then, all patients will be followed for survival status.

Any anticancer treatment administered after the last study treatment administration, if any, will be at the discretion of the investigator and recorded in the Case Report Form until the study cut-off date or death, whichever comes first.

8.15 TREATMENT ACCOUNTABILITY AND COMPLIANCE

The investigator or pharmacist will inventory and acknowledge receipt of all shipments of the investigational product (aflibercept/placebo). The investigational product (aflibercept/placebo) must be kept in a locked area with restricted access. The investigational product (aflibercept/placebo) must be stored and handled in accordance with the manufacturer's instructions.

Administration of the study drugs will be supervised by the investigator or subinvestigator.

The person responsible for drug dispensing is required to maintain adequate records of all study drugs. The labels of the aflibercept/placebo vials administered or dispensed to patients must be completed (patient number, and date of infusion, respectively). The packaging reference of investigational drug (aflibercept/placebo) and the lot number of docetaxel must be recorded in the CRF/drug accountability form, as well as the total number of vials per cycle.

The person responsible for drug administration to the patient will record precisely the date when the drug is administered to the patient. Interruption of the 1-hour aflibercept/placebo infusion or 1-hour infusion of docetaxel along with reason for the interruption will be recorded in the CRF.

The study monitor will periodically check the supplies of investigational product held by the investigator or pharmacist to verify accountability of all investigational product used. All unused investigational product and all medication containers will be returned to the Sponsor unless other arrangements have been approved by the Sponsor. The Sponsor will verify that a final report of drug accountability to the unit dose level is prepared and maintained in the investigator study file.

9 ASSESSMENT OF INVESTIGATIONAL PRODUCT

9.1 EFFICACY

9.1.1 Primary criteria

The primary endpoint is overall survival (OS) defined as the time interval from the date of randomization to the date of death due to any cause. In the absence of confirmation of death, survival time will be censored at the last date the patient is known to be alive.

Clinical examination will be performed every 3 weeks before each treatment cycle during the treatment period and then overall survival will be evaluated by collecting status information at 2 month intervals until death or the study cut-off date, whichever comes first.

9.1.2 Secondary criteria

9.1.2.1 Progression Free Survival

Progression Free Survival (PFS) will be defined as the time interval between the date of randomization and the time of occurrence of the first documented among the following events:

- Radiological tumor progression on CT scan and /or MRI
- Death due to any cause

9.1.2.2 Response rate (RR)

RR will be evaluated according to the RECIST criteria [24], see [Appendix B](#).

Chest, brain and abdominal CT or MRI scan, and other exams as clinically indicated to assess target and non-target lesions will be performed to assess disease status at baseline and will be repeated (same method as used as baseline) at the end of every 6 weeks or even-numbered cycle (i.e., Cycles 2, 4, 6, 8, etc.), and whenever disease progression is suspected (e.g., symptomatic deterioration). Bone scans will be performed at baseline and will be repeated: i) for patients with no bone lesions at baseline, whenever new or worsening bone symptoms occur; ii) for patients presenting bone lesions, every 3 months and whenever progression is suspected. These tests will be repeated to confirm a partial or complete response (at least 4 weeks after initial documentation of response) and at the end of study treatment. The investigator at the site will be responsible for the assessment and collection of the radiographic information in compliance with the schedule of evaluations presented in this protocol. During the follow-up period, patients who went off study treatment prior to documented disease progression will be evaluated by CT scan/MRI for tumor progression every 6 weeks from End of Study Treatment until disease progression or study cutoff date whichever comes first.

The overall response rate will be defined as the proportion of patients with confirmed RECIST-defined complete response (CR) or partial response (PR) relative to the total number of patients in the analysis population considered.

9.1.2.3 Health Related Quality of life (HRQL)

Rationale

In locally advanced or metastatic NSCLC where life expectancy remains short whatever the treatment, the impact the treatment have on health related quality of life (HRQL) takes a great importance to support both survival and response outcomes.

The hypothesis is that the key determinant of HRQL in this patient population is disease progression. Therefore HRQL evaluation will be performed using the Lung Cancer Symptom Scale (LCSS) questionnaire (see [Appendix G](#)), a validated instrument specific to lung cancer and focusing on lung cancer symptoms.

Instrument description

The LCSS scale, developed by Kris, Hollen and Gralla [28], consists of nine items, six measuring major symptoms for lung malignancies (appetite, fatigue, cough, dyspnea, hemoptysis and pain), and 3 summation items related to total symptomatic distress, activity status and overall quality of life. The recall period is the past day based on the rationale that the scale items represent states or conditions that fluctuate quickly [32]. The LCSS has a large published literature documenting its good psychometric properties [28, 30, 31].

Visual analogue scales with 100-mm lines are used to measure the intensity of patient responses. Each item is given an individual score equal to the length of the line marked by the patient, with zero corresponding to the lowest (best) rating and 100 representing the highest (worst) rating (ie. the highest degree of severity of symptoms). An average of the aggregate score of all the 9 items is used as total score. In addition, the mean of the 6 major lung cancer symptoms (appetite, fatigue, cough, dyspnea, hemoptysis and pain) provides an additional index called the Average Symptom Burden Index (ASBI).

HRQL will be assessed in all countries in the local languages. The LCSS has been translated into at least 45 languages and sub-languages. Questionnaires will be self-administered. It takes 5-8 min to complete initially for demonstration of the VAS, and only 3-5 min for subsequent administrations.

Timing of assessment

Baseline assessment should be obtained from all patients. The questionnaire will be administered within 8 days prior to randomization or at randomization, but in any case before the first administration of the study drug and before premedication. While on treatment, assessments

should occur on cycle 2 and 4, before administration of the next study treatment, and at the end of treatment visit.

Questionnaires should be filled in by the patient at the center prior to informing the patient about disease evolution and before the next study treatment. It is mandatory that a key person (e.g., research nurse) at each center be responsible for questionnaire data collection, in order to optimize compliance of the patient and to ensure completeness of the data.

9.1.2.4 Pharmacokinetics and Immunogenicity of aflibercept

See details in section 9.3.

9.2 SAFETY

Adverse event data will be collected by reporting at specified intervals throughout the study (see Sections 1.2 and 10). The results of weight, performance status and blood pressure will be recorded in the case report forms. Laboratory safety studies will be carried out according to standard operating procedures by the local laboratory. Abnormal, clinically significant results will be verified to rule out laboratory error. Persistent relevant abnormal values must be followed up until the cause is determined or until they return to the baseline value.

The study-specific and general safety criteria are developed in Section 10.1.

9.3 PHARMACOKINETICS AND IMMUNOGENICITY

Immunogenicity assessment is intended to be performed for all patients, with pharmacokinetic sampling performed at the same time points.

For all patients, one additional peak level of free and bound aflibercept will be measured at C1, immediately after the first aflibercept/placebo administration.

In order to permit a better understanding of the mechanism of action of aflibercept, free and bound aflibercept will be measured by validated ELISA methods. The ratio free aflibercept/bound aflibercept will be also estimated as an indicator of the presence of circulating endogenous VEGF.

Immunogenicity will be explored after exposure to aflibercept.

Baseline serum endogenous VEGF level will be determined in selected centers, in approximately 100 evaluable patients. Therefore, considering the double blind study context, samples have to be collected in 200 patients.

9.3.1 Sampling time and size

Plasma free and bound aflibercept concentrations will be estimated (see Figure 4 and Figure 5) before the start and at the end of the first infusion (cycle 1), immediately prior to third infusion (end of cycle 2), immediately prior to fifth infusion (end of cycle 4), then every other cycle, and

then approximately 30 days and finally approximately 90 days after last dose of aflibercept/placebo administration.

Serum for measurement of endogenous VEGF level will be collected at baseline before the start of the first infusion in selected centers (cycle 1).

Serum for measurement of anti-aflibercept antibody levels will be collected on cycle 1, before start of first infusion, then every other cycle, and then approximately 30 days and finally approximately 90 days after last dose of aflibercept/placebo administration, and in the circumstances of specific adverse events (see section 1.2. and section 8.5).

Figure 4 - PK sampling during study treatment

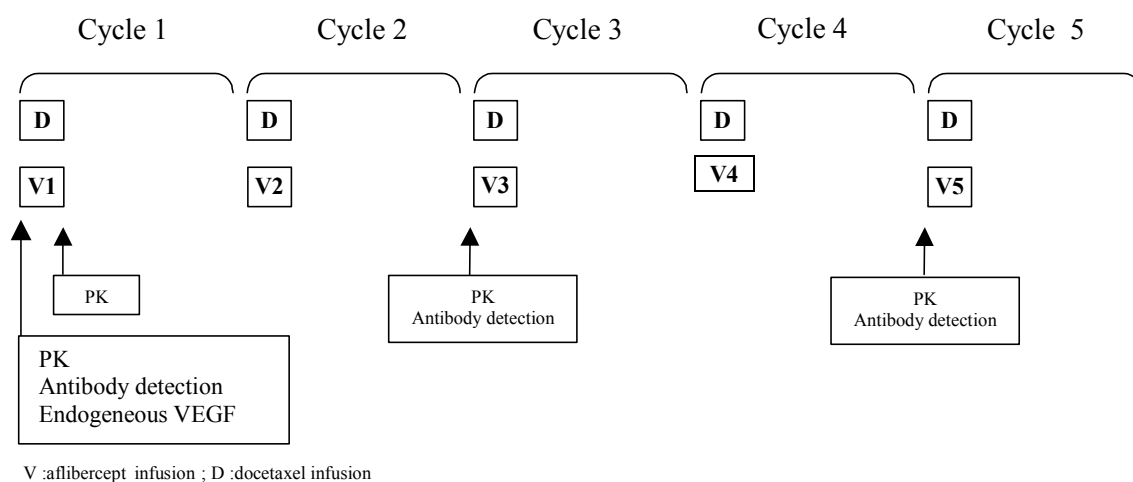
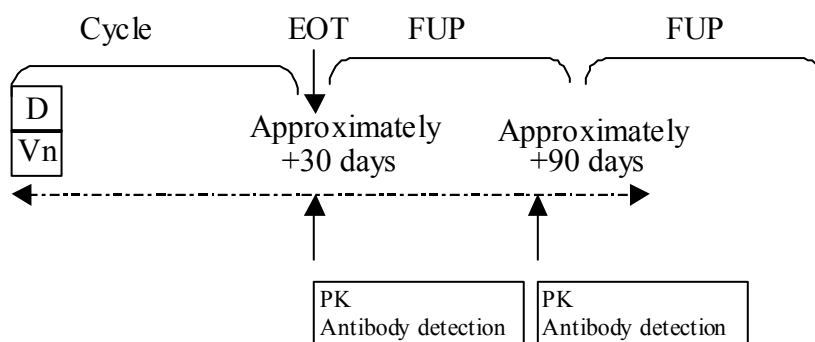


Figure 5 - PK sampling during follow-up



V :aflibercept infusion ; D :docetaxel infusion

9.3.2 PK handling procedure

Vacutainer tubes will be used to collect 4 mL of whole blood from patients for plasma preparation for the determination of circulating free aflibercept, for VEGF : aflibercept complex levels and for

pharmacokinetic analysis at the sampling times described in Section 9.3.1 above. Refer to [Appendix C](#) for detailed pharmacokinetic blood sample collection, handling, and shipping procedures. For 6 cycles of treatment, a total of 6 samples/patient and 24 mL of whole blood will be collected for these evaluations.

Blood samples should not be taken at the infusion site.

Red-top vacutainer tubes will be used to collect 4 mL of whole blood from patients for serum preparation for the analysis of anti-aflibercept antibody levels. Detailed anti-aflibercept antibody blood sample collection, handling, and shipping procedures are provided in [Appendix C](#). For 6 cycles of treatment a total of 5 samples and 20 mL of whole blood will be collected for these evaluations.

The analysis of free and bound aflibercept and anti-aflibercept antibody levels will be performed by Regeneron Pharmaceuticals, Inc. [REDACTED].

ACD or CPD tubes will be used to collect blood samples for the analysis of free endogenous VEGF levels at baseline in selected centers. Refer to [Appendix C](#) for detailed pharmacokinetic blood sample collection, handling, and shipping procedures. A total of one sample per patient and 4 mL of whole blood will be collected for this evaluation. All specimens must be stored frozen at -20°C until shipped to sanofi-aventis.

The plasma free VEGF concentrations will be determined at sanofi-aventis, GMPK Alfortville under the responsibility of [REDACTED]

9.3.3 Bioanalytical methods

Free aflibercept concentrations in plasma will be measured by a validated ELISA method. The ELISA microplates are coated with human VEGF, which specifically binds functional aflibercept, while the detection antibody is directed against the receptor domains of the aflibercept. By requiring a vacant VEGF binding site, this ELISA specifically measures free aflibercept; it does not detect the VEGF:aflibercept complex.

The assay of VEGF:aflibercept complex will also be measured by a validated ELISA method. The ELISA captures the complex with an antibody selective for VEGF coated on a plate. The captured complex is detected with an antibody selective for the receptor domains of aflibercept.

Human antibodies selective for aflibercept will be detected with an enzyme-linked immunosorbent assay that uses microplates coated with extracellular receptor domains of aflibercept. Immobilized receptor domain-antibody complexes are detected using peroxidase conjugated mouse anti-human IgG, F(ab')₂ fragment specific antibody. The calibration standard is a mouse monoclonal antibody specific for the R1 domain of aflibercept that is detected with a peroxidase-conjugated goat anti-mouse IgG Fc fragment-specific antibody.

Free endogenous VEGF will be detected by sandwich enzyme immunoassay using an anti-VEGF monoclonal antibody coated on the plates. VEGF is detected using an enzyme-linked polyclonal antibody specific for free VEGF. The limit of detection is 15 pg/mL.

9.4 MEASURES TO PROTECT BLINDING OF THIS TRIAL

Please also refer to Section 8.10: access to the randomization code during the study.

Patients, investigators, and other persons responsible for study conduct and data analyses will be blinded to treatment assignment.

All samples for PK evaluation will be assessed blindly (treatment arm and timing) and results will not be communicated to the study staff.

During the course of the study an external statistician (independent from the sponsor) will perform unblinded safety **and interim efficacy (IA)** analyses for the purpose of the DMC data review. Access to these data and analyses will be restricted to the DMC members, unless decision is made by the DMC to release results to the Steering Committee. Conditions to release unblinded results to the Steering Committee and processes to protect the integrity of the study are described in the DMC charter.

In case of a SAE that is unexpected and reasonably associated with the use of the investigational product the code will be broken by the safety officer in charge of the study for the purpose regulatory reporting.

In case of an AE, the code can be broken at the investigator initiative only in exceptional circumstances, when knowledge of the investigational product is essential for treating the patient. The investigator or authorized person should follow the procedures outlined in Section 8.10.

10 PATIENT SAFETY

10.1 SAFETY ENDPOINTS ASSESSED IN THIS TRIAL

The NCI CTCAE v.3.0 will be used in this study ([Appendix E](#)) to grade clinical adverse events and laboratory data.

Information on the following parameters will be collected by the investigator and reported in the CRF:

- Clinical examination, including body weight, blood pressure and ECOG PS.
- Laboratory data
 - Complete blood count and clinical chemistry (see study flowchart footnotes ‘e’ and ‘g’ in [Section 1.2](#))
 - Other tests as clinically indicated (e.g., 12-lead ECG)
- Adverse events and serious adverse events
- Concomitant medications and treatments
- Reason of treatment withdrawal and date of last administration
- Adverse events encountered before the start of aflibercept treatment will be summarized separately. Pre-treatment and treatment-emergent adverse events will be summarized with respect to the type as assessed by the Medical Dictionary for regulatory activities (current version or immediate previous version), frequency, cycle, severity according to the NCI CTCAE v.3.0., seriousness, and relatedness. Laboratory abnormalities will be assessed according to the NCI CTCAE v.3.0.

Please also refer to [Section 11](#).

10.2 SAFETY INSTRUCTIONS

10.2.1 Physical examination

Physical examination will include, but not limited to the examination of major body systems:

- Blood pressure
- Height (screening only), body weight
- ECOG PS ([Appendix D](#))

If abnormal findings emerge or worsen from the baseline assessment, then the adverse event page of the CRF should be completed for these findings. If a finding meets the criteria for a serious adverse event, then the appropriate procedures for reporting such events should be followed as described in [Section 10.4](#). Height will be recorded at baseline only. Body weight will be assessed

prior to the start of each treatment cycle. ECOG PS will be recorded prior to the start of each treatment cycle and every 2 months during follow-up period until the documentation of disease progression.

Every attempt should be made to have the same study personnel to perform the assessment throughout the study for any given patient for consistency of grading.

10.2.2 Laboratory variables

Hematology panel, blood chemistry profile as well as urine analysis will be performed by a local laboratory. Baseline results must be available for eligibility determination. At the start of each new treatment cycle, results must be available prior to treating the patient with the study drug.

10.3 ADVERSE EVENTS MONITORING

All events will be managed and reported in compliance with all applicable regulations, and included in the final clinical study report.

10.4 DEFINITIONS OF ADVERSE EVENT (AE) AND SERIOUS ADVERSE EVENT (SAE)

An **Adverse Event** is any untoward medical occurrence in a patient or clinical investigation patient administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment.

A priori, efficacy endpoints as specified in the protocol will not be considered as AEs except if, because of the course or severity or any other features of such events, the Investigator, according to his/her best medical judgment, considers these events as exceptional in this medical condition.

A **Serious Adverse Event** is any untoward medical occurrence that at any dose:

- Results in death or;
- Is life-threatening or;

Note: The term "life-threatening" in the definition of "serious" refers to an event in which the patient was at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe.

- Requires inpatient hospitalization or prolongation of existing hospitalization or;
- Results in persistent or significant disability/incapacity or;
- Is a congenital anomaly/birth defect;
- Is a medically important event:
Medical and scientific judgment should be exercised in deciding whether expedited reporting is appropriate in other situations, such as important medical events that may not be immediately life-threatening or result in death or hospitalization but may jeopardize the

patient or may require intervention to prevent one of the other outcomes listed in the definition above.

Note: Examples of such events are intensive treatment in an emergency room or at home for allergic bronchospasm, blood dyscrasias, convulsions or asymptomatic ALT increase ≥ 10 ULN that does not result in hospitalization, or development of drug dependency or drug abuse.

10.5 OBLIGATION OF THE INVESTIGATOR REGARDING SAFETY REPORTING

10.5.1 Adverse Events

The period of safety observation starts from the time the patient gives informed consent.

All Adverse Events regardless of seriousness or relationship to Investigational Product, spanning from the signing of the informed consent form until 30 days after the last study treatment administration, are to be recorded on the corresponding page(s) included in the Case Report Form.

During baseline period (i.e. even in the absence of any administration of study treatments), all signs and symptoms will be recorded as adverse event only if they are still present at the time of first study drug administration or if they are serious. They are to be recorded on the corresponding page(s) included in cycle 1 in the Case Report Form, except if they are serious (in this case they have to be reported on the baseline adverse event page).

During the treatment period (i.e. until 30 days after the last administration of study drugs), all adverse events, regardless of seriousness or relationship to study treatments, are to be recorded on the corresponding page(s) included in the Case Report Form.

For follow-up period, refer to Section [10.5.3](#).

Whenever possible, symptoms should be grouped as a single syndrome or diagnosis. The Investigator should specify the date of onset, intensity, action taken with respect to study treatments, corrective treatment/therapy given, additional investigations performed, outcome and his/her opinion as to whether there is a reasonable possibility that the Adverse Event was caused by study drugs.

Blood pressure or ECG abnormalities are to be recorded as Adverse Events only if they are medically relevant: symptomatic, requiring corrective treatment, leading to discontinuation/dose delay/dose reduction/dose interruption and/or fulfilling a seriousness criterion.

Laboratory abnormalities are to be recorded as Adverse Events only if they lead to discontinuation/dose delay/dose reduction/dose interruption and/or are fulfilling a seriousness criterion.

10.5.2 Serious Adverse Events

In the case of a Serious Adverse Event the Investigator must immediately:

- SEND (within 1 working day) by email (automatic sending activated by ticking “save complete” (for the initial report) or “save” (for follow-up reports) the electronic Adverse Event/Serious Adverse Event notification (html format) to the representatives of the Monitoring Team in the Clinical Research Unit (Clinical Safety Officer and Clinical Research Associates) who then forwards to Global Pharmacovigilance and Epidemiology and the Clinical Study Director. Before sending, please verify that relationship to study drugs has been indicated on the AE form and treatment number is provided on the SAE complementary form.
- APPROVE the Adverse Event/Serious Adverse Event page in the electronic CRF immediately after automatic sending of the electronic Serious Adverse Event notification by entering, for a second time, “username and password”;
- If the mail connection is not functional or in case of any technical issue, SEND (within 1 working day, preferably by fax) the signed and dated corresponding page(s) in the Case Report Form to the representative of the Monitoring Team whose name, address and fax number appear on the Clinical Trial Protocol;
- ATTACH AND FAX photocopy of all examinations carried out and the dates on which these examinations were performed. Care should be taken to ensure that the patient’s identity is protected and the patient’s identifiers in the Clinical Trial are properly mentioned on any copy of source document provided to the Sponsor. For laboratory results, include the laboratory normal ranges;
- Follow-up of any Serious Adverse Event that is fatal or life threatening should be provided within one additional calendar week.

10.5.3 Follow-up

The Investigator should take all appropriate measures to ensure the safety of the patients, notably he/she should follow up the outcome of any related Adverse Events (clinical signs, laboratory values or other, etc.) until the return to normal or consolidation of the patient's condition.

In case of any Serious Adverse Event, the patient must be followed up until clinical recovery is complete and laboratory results have returned to normal, or until progression has been stabilized. This may imply that follow-up will continue after the patient has left the Clinical Trial and that additional investigations may be requested by the Monitoring Team.

In case of any Adverse events or Serious Adverse Event brought to the attention of the Investigator at any time after cessation of study treatment and considered by him/her to be caused by the study treatment with a reasonable possibility, should be reported to the Monitoring Team.

10.6 PREGNANCY

Pregnancy will be recorded as an AE in all cases. It will be qualified as an SAE only if it fulfills SAE criteria.

In the event of pregnancy, Investigational Product should be discontinued and the Sponsor informed immediately (i.e. within 1 working day), even not fulfilling a seriousness criterion, using the AE form together with the SAE complementary form to be sent to the representative of the

monitoring team whose name, address and fax number appear on the clinical trial protocol.

Follow-up of the pregnancy will be mandatory until the outcome has been determined.

10.7 OVERDOSAGE

No cases of aflibercept overdose, defined as any administered dose at least 1 dose level higher than the highest explored dose level, have been reported to date. The highest doses administered of aflibercept are 7 mg/kg IV every 2 weeks and 9 mg/kg IV every 3 weeks.

For safety reporting purposes, in this protocol, overdose of docetaxel is defined as $> 100 \text{ mg/m}^2$ every 3 weeks.

10.8 OBLIGATIONS OF THE SPONSOR

During the course of the study, the Sponsor will report in an expedited manner all SAEs that are both unexpected and at least reasonably related to the study treatment regimen, to the Authorities, IECs / IRBs as appropriate and to the Investigators. The determination of the expectedness for SAEs, for regulatory reporting purposes, will be defined by the current Investigator's brochures of aflibercept and of docetaxel (Taxotere® Injection concentrate) in force at the moment of the event occurrence.

In addition, the Sponsor may report in an expedited manner all SAEs that are expected and at least reasonably related to the study treatment to the Authorities, according to local regulations.

In this study, disease progression, or lack of efficacy, will not be considered unexpected, unless the course, intensity or other specific features are such that the Investigator, according to his/her best medical judgment, considers these events as exceptional in the context of this medical condition. Any other SAE not consistent with SAEs listed in the Investigator's Brochures of aflibercept and of docetaxel will be considered as unexpected.

The Sponsor will report all safety observations made during the conduct of the trial in the clinical study report.

11 HANDLING OF PATIENT DEFINITIVE TREATMENT DISCONTINUATION AND OF PATIENT STUDY DISCONTINUATION

11.1 DEFINITIVE TREATMENT DISCONTINUATION

11.1.1 List of criteria for definitive treatment discontinuation

The patients may withdraw from treatment (either aflibercept/placebo or docetaxel) under the following circumstances but will continue to be assessed and followed in the study unless the patient refuses:

- The patients may withdraw from study treatment if they decide to do so, at any time and irrespective of the reason (consent's withdrawal) or at the request of their legally authorized representative. "Legally authorized representative" means an individual or judicial or other body authorized under applicable law to consent on behalf of a prospective patient to the patient's participation in the procedure(s) involved in the research.
- If, in the investigator's opinion, continuation of the study treatment would be detrimental to the patient's well being, such as:
 - Disease progression as defined in section 9.1.2.2,
 - Unacceptable adverse event(s) not manageable by symptomatic therapy, dose delay or dose modification (see Section 8.4 and Section 8.5),
 - Intercurrent illness that prevents further administration of study treatment,
 - Non compliance to the study protocol or logistic consideration.
- Patient is lost to follow-up
- Of note, any treatment unblinding by the Investigator will lead to permanent discontinuation of the investigational product.

In all cases, the reason for and date of withdrawal must be recorded in the CRF and in the patient's medical records. The patient must be followed up to establish whether the reason was an adverse event, and, if so, this must be reported in accordance with the procedures in Section 10.

11.1.2 Handling of patients after definitive treatment discontinuation

All definitive treatment discontinuation should be recorded by the Investigator in the appropriate pages when considered as confirmed.

Patients will be followed up according to the study procedures as specified in this protocol up to the scheduled date of study completion, or up to recovery or stabilization of a followed-up AE, whichever comes last.

If possible, and after the permanent discontinuation of treatment, the patients will be assessed using the procedure normally planned for the last dosing day with the Investigational Product, including antibody and pharmacokinetics measurement.

11.2 PROCEDURE FOR WITHDRAWAL OF PATIENTS FROM STUDY FOLLOW-UP SCHEDULE

The patients may withdraw from the study follow-up schedule, before study completion if they decide to do so, at any time and irrespective of the reason, or this may be the Investigator's decision.

All study withdrawals should be recorded by the Investigator in the appropriate CRF pages and in the patient's medical records when considered as confirmed (at least date of withdrawal and reason for).

If possible, the patients are assessed using the procedure normally planned for the end-of-study visit as described in Section [12.1.4](#).

The investigator must make every effort to contact patients lost to follow-up. Attempts to contact such patients must be documented in the patient's records (e.g., times and dates of attempted telephone contact, receipt for sending a registered letter). Patients who did not complete the study and for whom no endpoint data are available will be considered as lost to follow-up. Patients lost to follow-up will be censored for survival analysis at the time they were last known to be alive (see primary endpoint definition in section [13.2.2.1](#)).

11.3 CONSEQUENCE

Patients who have been withdrawn from the study cannot be reincluded in the study. Their inclusion and treatment number must not be reused.

12 STUDY PROCEDURES

12.1 VISIT SCHEDULE

12.1.1 Pre-treatment evaluation (baseline / screening visit)

- The **written informed consent** will have to be signed by the patient before any protocol specific procedures.

The following examinations will be performed within 21 days prior to study randomization:

- **Demographics:** age (date of birth), gender, and race
- **Medical, surgical and oncological history** including significant prior and concurrent illnesses, existing signs and symptoms, primary diagnosis and prior antitumor therapy.
- **Prior medications** will be recorded from 21 days prior to the start of the study treatment.
- **Tumor assessment** : Chest, brain and abdominal CT or MRI and bone scan and all other exams as clinically indicated, to assess all TARGET or NON TARGET lesions (measurable and non measurable).

When available, spiral CT acquisition should be done. Slice thickness should be adapted to the anatomical area and presumed size of the lesions. Slice thickness of 5 to 8 mm should be favored rather than 10 mm, especially during spiral acquisition. If limitations appear in volume acquisition, it is encouraged to choose a 1.5 pitch and thin slices, rather than a 1 pitch with thick slices. A centimeter scale should appear on films.

The following examinations will be performed within 8 days prior to study randomization:

- **Clinical examination** including major body systems exam, height and weight, ECOG PS, and blood pressure.
- **Inclusion/Exclusion criteria:** each potential patient will be examined before the start of the study to determine his/her eligibility for participation.
- **Laboratory safety assessments:**
 - **Hematology:** WBC, ANC, hemoglobin, platelet count.
 - **Blood Chemistry** sodium, potassium, calcium, phosphorus, BUN, magnesium, creatinine, creatinine clearance (calculated with Cockcroft-Gault formula) [29, [Appendix A](#)], total protein, albumin, SGOT (AST), SGPT (ALT), alkaline phosphatase, total bilirubin, glucose.
 - **Coagulation tests:** prothrombin time expressed as INR in patients under anticoagulants
 - **Urinalysis:** dipstick (WBCs, RBCs) and UPCR on morning urine spot or 24 hour urine collection to assess 24-h urine protein rate.
 - **Pregnancy test** in women of reproductive potential: serum or urine β -hCG must be measured within 8 days of first study treatment administration.

- **12-lead ECG.**
- **LCSS questionnaire.**

12.1.2 Randomization

Randomization will take place once the consented patient has completed all the necessary screening procedures and is deemed eligible for study entry by the investigator or designee. All eligible patients must be randomized by contacting the IVRS (see section 8.7).

The results of the screening examinations will be recorded in each randomized patient's CRF. Source documentation to support the screening results must be maintained in the patient's medical record. Treatment should begin within 3 days after randomization.

12.1.3 During study treatment period

The Study treatment period begins when the patient receives the initial dose of study drugs (Cycle 1 Day 1). Each cycle consists of 21 days and assessments are scheduled on an every 3-week basis (prior to dosing of each cycle) but may be repeated more often, as clinically indicated. Cycle length may be extended if additional time is required for resolution of study drug-related toxicities or other adverse events, but cycle shortening to less than 19 days is not permitted. A maximum 2-week delay for resolution of study-drug related toxicities is allowed. Beyond this, the Sponsor should be contacted for guidance.

Each randomized patient will be treated with aflibercept/placebo every 3 weeks.

In- or outpatient status will be recorded in the CRF.

The following examinations will be performed every cycle:

- **Clinical examination:** (before each cycle): ECOG PS, weight, examination of major body systems, blood pressure. Blood pressure will be measured and recorded in the CRF at least once between visits during week 2 of each cycle. If BP >150/100 mmHg or diastolic pressure increase > 20 mmHg from baseline, initiate or modify antihypertensive drug therapy as per recommendation in Section 8.5.1.
- **Assessment of concomitant medications.**
- **All adverse events** will be assessed before each cycle.
- **Laboratory safety studies:**
 - **Hematology:** WBC, ANC, hemoglobin, platelet count. In case of neutopenia grade 3-4 repeat ANC every 2 days until recovery to grade ≤ 1. In case of fever during any cycle blood count will be performed.
 - **Blood Chemistry:** sodium, potassium, calcium, phosphorus, BUN, magnesium, creatinine, creatinine clearance (calculated with Cockcroft-Gault formula), total protein, albumin, SGOT (AST), SGPT (ALT), alkaline phosphatase, total bilirubin, glucose.

- **Coagulation tests:** prothrombin time expressed as INR in patients under anticoagulants
- **Urinalysis:** dipstick (WBCs, RBCs) and UPCR on morning urine spot:
 - if UPCR is >1 then 24-hour urine collection must be performed to assess the urine protein rate, as well as a urine protein electrophoresis,
 - if proteinuria is associated with hematuria then LDH, haptoglobin, schistocytes and orosomucoid and anti-aflibercept antibodies will be measured in blood.

(See section 8.5.1)

- **Other investigations if clinically indicated.**

The following examinations will be performed every 2 cycles:

- **Tumor assessment**, every 6 weeks from first cycle (except for bone scan that will be performed: i) for patients with no bone lesions at baseline, whenever new or worsening bone symptoms occur; ii) for patients presenting bone lesions every 3 months or at any time in case of clinical suspicion of bone progression) to follow target and /or not target lesion present at baseline. To ensure comparability, the imaging should be performed using identical techniques throughout the study period (i.e., scans performed immediately following bolus contrast administration using a standard volume of contrast, the identical contrast agent, and preferably the same scanner).
To be performed **every 6 weeks** (2 cycles) until disease progression, even if the patient discontinued study treatment before disease progression, or study cut-off date whichever comes first.
- **LCSS (cycles 2 and 4 only).**
- **Anti-aflibercept antibodies in all patients, just before the start of the first infusion of aflibercept/placebo, and then pre-dose every other cycle during treatment (at the end of cycles 2, 4 and so on).**
- **Blood samples for Pharmacokinetics** will be collected for all patients, just before the start and at the end of the first infusion aflibercept/placebo and then pre-dose every other cycle during treatment (at the end of cycles 2, 4 and so on).
- **In selected centers for all patients, blood samples for determination of endogenous VEGF level just before the start of the first infusion aflibercept/placebo.**

12.1.4 End of treatment

All patients must continue to be observed for at least 30 days after the final dose of study treatment (either aflibercept/placebo or docetaxel)

The following procedures should be performed approximately 30 days following the final dose of study treatment:

- **Clinical examination:**
ECOG PS, weight, examination of major body systems, including blood pressure.
- **Assessment of concomitant medications.**

- **Tumor assessment:** if not performed within the previous 6 weeks.
- **LCSS.**
- **Assessment of adverse events.**
- **Laboratory safety studies:**
 - **Hematology:** WBC, ANC, hemoglobin, platelet count.
 - **Blood Chemistry** sodium, potassium, calcium, phosphorus, BUN, magnesium, creatinine, creatinine clearance (calculated with Cockcroft-Gault formula), total protein, albumin, SGOT (AST), SGPT (ALT), alkaline phosphatase, total bilirubin, glucose.
 - **Coagulation tests:** prothrombin time expressed as INR in patients under anticoagulants
 - **Urinalysis:** dipstick (WBCs, RBCs) and UPCR on morning urine spot:
 - If UPCR is >1 then 24 hour urine collection must be performed to assess the urine protein rate, as well as a urine protein electrophoresis,
 - if proteinuria is associated with hematuria then LDH, haptoglobin, schistocytes and orosomucoid and anti-aflibercept antibodies will be measured in blood.

(See section [8.5.1](#))

- Other investigations if clinically indicated.

For all patients:

- **Blood samples for Pharmacokinetic:** approximately 30 days after the last infusion of aflibercept/placebo.
- **Anti-aflibercept antibody:** approximately 30 days after the last infusion of aflibercept/placebo.

12.1.5 Post treatment Follow-up period

The first follow up visit will occur approximately 90 days after the final dose of study treatment (i.e., 60 days after the end of treatment visit).

- **Survival status, ECOG PS.**
- **Clinical examination:**
ECOG PS, weight, examination of major body systems, including blood pressure.
- **Assessment of concomitant medications.**
- **Tumor assessment:** when disease progression is not documented at end of treatment, to be performed **every 6 weeks** until disease progression.
- **All adverse events related to study treatment or any SAE ongoing** at the end of the study treatment, or new related AE and related SAE which occur during the follow-up period will be recorded until recovery, or until progression has been stabilized.

- **Further anticancer therapy** if any, which is at the discretion of the investigator. No further antitumor therapy should be administered before disease progression unless the patient requests further antitumor therapy, or the investigator deems it necessary.

For all patients:

- **Blood samples for Pharmacokinetic:** approximately 90 days after the last infusion of aflibercept/placebo.
- **Anti-aflibercept antibody:** approximately 90 days after the last infusion of aflibercept/placebo.

Subsequent Follow up visits will continue there after every 2 months until death, or study cut-off, whichever comes first.

The followings will be performed:

- **Survival status; ECOG PS** (until disease progression is documented)
- **Assessment of medications** if correspond to treatment of related adverse events.
- **Tumor assessment:** every 6 weeks until disease progression is documented.
- **All adverse events related to study treatment or any SAE ongoing** at the end of the study treatment, or new related AE and related SAE which occur during the follow-up period will be recorded until recovery, or until progression has been stabilized.
- **Assessment of further anticancer therapy** if any, which is at the discretion of the investigator. No further antitumor therapy should be administered before disease progression unless the patient requests further antitumor therapy, or the investigator deems it necessary.

12.1.6 End of Study

All patients will be followed for survival status and further anticancer therapy every two months until death or the study cut-off date, whichever comes first.

12.2 DEFINITION OF SOURCE DATA

Source data includes all information in original records and certified copies of original records of clinical findings, observations, or other activities necessary for the reconstruction and evaluation of the trial. Source data are contained in source documents.

Source documents are original documents, data and records (e.g. hospital records, clinical and office charts, laboratory notes, memoranda, patient diaries or evaluation checklists, pharmacy dispensing records, recorded data from automated instruments, copies or transcripts certified after verification as being accurate and complete, microfiches, photographic negatives, microfilm or magnetic media, X-rays, patient files, and records kept at the pharmacy, at the laboratories, and at medical-technical departments) involved in the clinical study. Source documentation must be maintained to support information provided within a CRF.

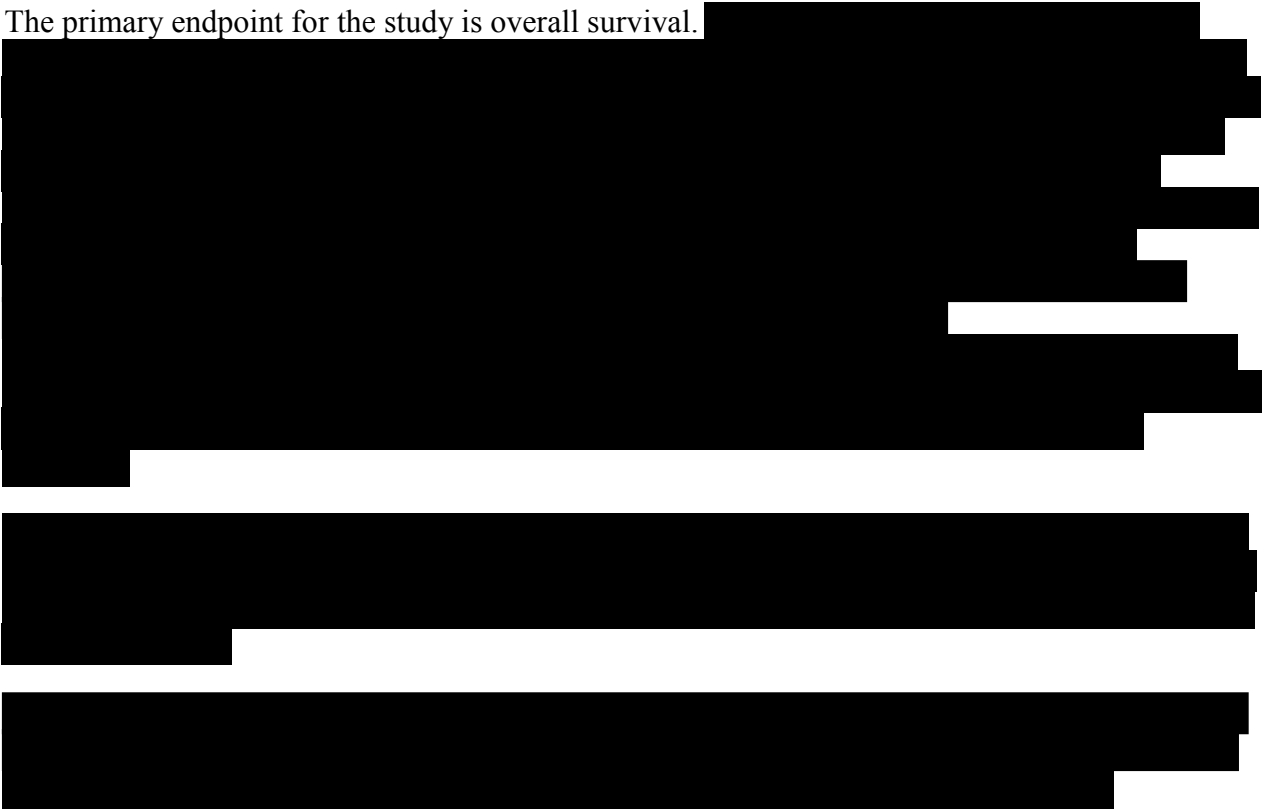
The results of certain examinations or evaluations recorded in the CRF may be considered to be source data (such as LCSS, ...).

13 STATISTICAL CONSIDERATIONS

The material of Section 13 of the Clinical Trial Protocol is the basis for the Statistical Analysis Plan (SAP) for the study. This plan may be revised during the study to accommodate Clinical Trial Protocol amendments and to make changes to adapt to unexpected issues in study execution and data that affect planned analyses. These revisions will be based on blinded review of the study and data, and a final plan will be issued at the latest 6 to 8 weeks before final database lock. **A final SAP will be issued prior to the database lock of the interim analysis.**

13.1 DETERMINATION OF SAMPLE SIZE

The primary endpoint for the study is overall survival.



13.2 ANALYSIS VARIABLES

13.2.1 Demographic and baseline characteristics

Standard demographic and baseline characteristics such as age, gender, race, height and weight will be summarized.

Other characteristics data at baseline that has potential influence on the primary outcome of the study will be examined by treatment group and compared using appropriate descriptive

techniques. These characteristics include ECOG Performance Status, cancer diagnosis, tumor involvement, and prior anticancer therapy.

Baseline efficacy and safety variables such as tumor assessment and major lab parameters as well as other prognostic variables will also be assessed.

13.2.2 Efficacy variables

13.2.2.1 Primary efficacy variable(s)

The primary efficacy endpoint is overall survival defined as the time interval from the date of randomization to the date of death due to any cause.

In the absence of confirmation of death, survival time will be censored at the last date the patient is known to be alive.

13.2.2.2 Secondary efficacy variable(s)

Main secondary endpoints and exploratory variables are:

Tumor progression free survival (PFS) defined as the time interval between the date of randomization and the date of either first RECIST-defined tumor progression or death due to any cause, whichever is earlier.

If death or progression is not observed, data on PFS will be censored at the earlier of the date of last tumor assessment without evidence of progression or the study cut-off date. The details of the censoring rules will be provided in the SAP.

Objective response rate (RR) defined as the proportion of patients with confirmed complete response (CR) or confirmed partial response (PR), defined by RECIST, relative to the total number of patients in the analysis population.

Health Related Quality of Life (HRQL), assessed by the Lung cancer symptom scale (LCSS) questionnaire, will be scored. The LCSS total score is defined as the mean of the 9 items of the scale (score between 0 to 100). In addition, the Average Symptom Burden Index (ASBI) will be calculated as the mean of the 6 major lung cancer symptoms (appetite, fatigue, cough, dyspnea, hemoptysis and pain). Each individual score, the ASBI and the total scores will be described at each visit as well as change from baseline scores. Specific data handling conventions and analysis will be documented in the Statistical Analysis Plan.

13.2.3 Safety variables

The safety variables include:

- **AE**
 - On-treatment period: On-treatment period is the period from the first dose to 30 days after the last dose.

- Treatment-emergent AEs (TEAEs): A TEAE is defined as an AE that is reported, during the on-treatment period defined above.
- Post-treatment AEs: A “post-treatment AE” is defined as an AE that developed, worsened or became serious after completion of the on-treatment period.
- **Discontinuation**
 - Treatment discontinuation and reasons.
 - Treatment discontinuation due to AEs.
- Blood pressure (SBP, DBP) and ECOG performance status
- **Major laboratory safety parameters**
 - Hematology: neutrophils, platelets, and hemoglobin.
 - Selected Blood chemistry: total bilirubin, alkaline phosphatase, SGOT (AST), and SGPT (ALT), Creatinine.
 - Renal function and urinalysis: Dipstick (WBC, RBC), Urinary Protein-to-creatinine Ratio (UPCR), and 24-hour protein.

13.2.4 Pharmacokinetic variables

The PK analysis will include population pharmacokinetics of IV Aflibercept and endogenous VEGF level.

13.3 ANALYSIS POPULATIONS

13.3.1 Efficacy populations

Intent-to-Treat (ITT) population: This population includes all randomized patients.

This population is the primary population for all efficacy parameters. All analyses using this population will be based on the treatment assigned by IVRS.

Evaluable Patient (EP) population for tumor response: This population consists of all randomized patients with measurable disease at study entry who have received at least one full cycle of treatment, with lung cancer previously treated with a platinum based chemotherapy, and with at least one post baseline tumor evaluation, except for early disease progression/death.

The evaluable patient population for tumor response will be used for the RR analysis (secondary population for RR analysis). All analyses using this population will be based on the treatment actually received.

Details of the major protocol deviations will be defined in the Statistical Analysis Plan.

13.3.2 Safety population

All-Treated (AT) population: The AT population is a subset of the ITT population that took at least one dose of study medication (aflibercept, placebo or docetaxel)

All analyses using this population will be based on the treatment actually received.

For patients who were not randomized but took aflibercept by mistake, the patients' safety data will be reported in the report but these patients will not be included in the AT population for any statistical summary.

13.3.3 Other analysis population

All treated patients will be used to collect samples to perform the immunogenicity analyses.

All treated patients will be used to collect samples to perform the complete PK-analyses.

The pharmacokinetic analysis will be performed based on the all treated population with evaluable blood samples (drawn at pre and post-dose cycle 1, at pre-dose cycle 3, cycle 5, then every other cycle, and then approximately 30 and 90 days after the last infusion of aflibercept/placebo) e.g. whom had a pre-dose and at least one post dose assessment performed. Peak analysis will be performed at cycle 1 and trough analyses will be performed at cycle 2, cycle 4 and every other cycle.

The analysis of anti- aflibercept antibody measurement will be performed based on the all treated population with evaluable blood samples (drawn at pre-dose cycle 1, cycle 3, then every other cycle, and then approximately 30 and 90 days after the last infusion of aflibercept/placebo) e.g. whom had a pre-dose and at least one post dose assessment performed.

13.3.4 Disposition of patients

Summary of each patient population will be presented with counts (n) and percentages (%).

Study drug discontinuation will be summarized by reasons for each treatment arm using the AT population. The discontinuation from the study will be summarized by treatment arm and overall using the ITT population.

13.4 STATISTICAL METHODS

For each efficacy variable, if a statistical test is performed, the test will be performed at an overall two-sided alpha level of 0.05.

In general, the descriptive summary for continuous data will include the number of non-missing observations (n), mean, SD (standard deviation), median, minimum and maximum. In addition, 25%-percentile and 75%-percentile may be provided if needed.

The descriptive summary for categorical variables will include counts (n) and percentages calculated in the related group. The denominator will be determined by the analysis population (ITT, evaluable, or AT) used for the summary. Non-evaluable outcome or missing data will be handled based on the data handling strategy in Section 13.5.

Descriptive summary of the time-to-event data will include the Kaplan-Meier method, median time-to-event and its 95% confidence intervals.

13.4.1 Demographic and baseline characteristics

Unless otherwise stated, demographic and baseline characteristics will be summarized using descriptive statistics using the ITT population. Analyses for the all-treated population or the evaluable patient population for tumor response population may be presented if the number of patients differs substantially from the ITT populations

13.4.2 Extent of study treatment exposure and compliance

The summary of extent of exposure will include number of cycles administered, duration of dosing (weeks), cumulative dose (mg/kg for aflibercept and mg/m² for docetaxel), dose intensity (mg/kg/week for aflibercept and mg/m²/week for docetaxel) and relative dose intensity (%) by treatment group.

Dose delays, reductions, and interruption will be also summarized by reasons.

Extent of exposure will be assessed in the All-Treated population.

13.4.3 Analyses of efficacy variables

13.4.3.1 Analysis of primary efficacy variable(s)

The primary analysis of the OS will be the comparison between the two treatments using the log-rank test procedure stratified by stratification factors specified at the time of randomization.

The estimates of the hazard ratio and corresponding 95% confidence interval will be provided using a Cox proportional hazard model stratified by the same stratification factors as those used for the log-rank test described above.

The survival curves will be estimated using Kaplan-Meier estimates. **The final OS analysis will be conducted when 687 deaths have been observed. One interim analysis of OS for the purpose of futility and efficacy is planned when approximately 344 deaths (50% of deaths) have occurred. Using a group sequential approach with an O'Brien Fleming α -spending function and an overall one-sided α level of 0.025, the one-sided α nominal significance to be used at the final analysis is 0.0245 (see details on interim analysis in section 13.6).**

All of these analyses will be performed in the ITT population.

13.4.3.2 Analyses of secondary efficacy variables

Secondary endpoints are defined as follows:

- **Progression Free Survival (PFS):** PFS will be analyzed with the Kaplan-Meier method and compared between 2 treatments using log-rank test stratified by stratification factors at randomization. The analyses will be performed in the ITT population.
- **Objective Response Rate (RR):** RR is will be compared between 2 treatments using CMH test adjusted for stratification factors at randomization. The analyses will be performed based on the EP population and ITT population.
- **Health Related Quality of Life (HRQL):** Each individual score, the Average Symptom Burden Index (ASBI) and the total scores will be described at each visit as well as change from baseline scores. Specific data handling conventions and analysis will be documented in the Statistical Analysis Plan.

13.4.3.3 Type-I error rate control

In order to protect the overall type-I error level, a step-down procedure will be used in the analyses of efficacy variables according to the following sequential order: (1) Primary efficacy variable (OS), (2) PFS, (3) RR and (4) QoL. At each step, the procedure will stop if the test at the current step does not meeting the pre-determined significance level.

This procedure is applicable both at interim and final analysis of Overall Survival. All secondary endpoints among the step-down procedure will be tested at the 2-sided 5% level, whatever the stage of the statistical analysis (interim or final).

13.4.4 Analyses of safety data

Analysis of adverse events, blood pressure, ECG, and laboratory data, as defined in Section [13.2.3](#), will be descriptive by treatment group in the AT population. Further details will be provided in the SAP.

13.4.4.1 Adverse Events

TEAEs will be summarized with the number and percentage of patients with AEs, classified by MedDRA preferred term and intensity as graded by the NCI CTCAE version 3.0.

SAEs and treatment-emergent SAEs will be summarized using MedDRA classification up to the preferred term. SAEs will be also presented by worst NCI grade by patients.

TEAEs will be analyzed regardless of their relationship to the study drug. However, summaries will be done for specific subsets of AEs, such as those potentially related to study drug.

13.4.4.2 Laboratory safety data

Hematological toxicities will be assessed from laboratory parameters. Worst NCI CTCAE grades of leukopenia, neutropenia, thrombocytopenia, and anemia will be determined.

Qualitative and quantitative results will be summarized for hematological toxicities. Qualitative data (worst NCI CTCAE grade) will be summarized by cycle and by patient.

Biochemistry and urinalysis will be analyzed using the worst NCI CTCAE grade, whenever applicable (laboratory normal ranges, otherwise) calculated from laboratory values. Specific attention will be given to incidence of proteinuria and impairment of renal function.

13.4.4.3 Vital signs

By visit descriptive analyses of vital signs will be provided for observed values and change from baseline.

13.4.5 Analyses of pharmacokinetic and pharmacodynamic variables

The PK analysis will include population pharmacokinetics of IV aflibercept with its associated interpatient variability. Concentrations of free aflibercept, VEGF:aflibercept complex, and the ratio free aflibercept to VEGF:aflibercept complex levels will be estimated. Levels of circulating endogenous VEGF will be assessed as well

The population pharmacokinetics of free and bound aflibercept have been estimated from the Phase I studies in which intensive blood sampling protocols were used. The population estimates from this analysis provided a prior distribution from which individual Bayesian estimates of the pharmacokinetic parameters for each patient in this study will be derived.

13.4.6 Analyses of anti-aflibercept antibodies measurement

Anti- aflibercept antibody levels will be categorized as negative and positive. A patient will be considered to have positive anti-body levels if antibodies were detected above the quantification limits.

13.5 DATA HANDLING CONVENTIONS

Some general rules of data handling conventions are listed below. Further details will be provided in the SAP.

- Baseline values: In general, the baseline value of an analysis variable is defined the last observation before the first dose of the treatment. A mean value can be considered for certain variables if multiple observations are collected at baseline.
- Missing data: In general, no imputation is planned for missing data. The following approaches are default methods for missing data handling. Some exploratory analyses can be planned with different strategies for treating missing outcomes. The details will be provided in the SAP.

- Binary data: When a proportion is calculated for a binary variable, the denominator is based on the total number of patients in the analysis population used for the summary. There can be 3 observations: Yes, No, and Non-evaluable (or missing). For the patients with non-evaluable outcomes, the default rule is that the patients will be treated as “no events”.
- Continuous data: The analyses and summaries for variables with continuous scales will be based on observed data only. However, the number of patients with missing observations will be provided.
- Time-to-event data: Missing outcomes due to different reasons will be handled using different censoring rules. The censoring rules are specified as part of the definition of the analysis variables.
- Analysis windows: For summaries about AE, exposure and tumor assessments, Cycles based on CRF data will be used. For certain other measurements such as blood pressure, a window defined around the date of treatment may be considered. Details will be provided in the SAP.

13.6 INTERIM ANALYSIS

The objective of the interim analysis is to provide a methodological rationale and decision rules to the members of the DMC to either recommend to continue the study as planned or to stop the study earlier either because of already demonstrated efficacy or futility based on the pre-specified boundaries. Further details will be provided in the DMC charter.

The interim analysis of OS to stop for futility or overwhelming efficacy is planned when approximately 344 OS events (50% information fraction) have occurred. O’Brien-Fleming α -spending function will be used for the interim analysis of efficacy. Under this proposal, the total number of events required will be 687 with the interim analysis being done when approximately 344 OS events (50% information fraction) have occurred. For the boundary of stopping the trial for overwhelming efficacy in favor of aflibercept + docetaxel, the nominal one-sided significance level for efficacy will be 0.0015 at the time of the interim analysis, that is, the p-value of the one-sided log-rank test needs to be < 0.0015 at the time of interim analysis. Alternately, the positive boundary will be crossed if the hazard ratio at the interim is < 0.73 .

The futility boundary based on Gamma (-6) β -spending function will be crossed if the hazard ratio ≥ 1.03 in favor of the docetaxel + placebo arm. The conditional power under the alternative hypothesis at the time of interim analysis will be 0.2170.

14 ETHICAL AND REGULATORY STANDARDS

14.1 ETHICAL PRINCIPLES

This Clinical Trial will be conducted in accordance with the principles laid down by the 18th World Medical Assembly (Helsinki, 1964) and all applicable amendments laid down by the World Medical Assemblies, and the ICH guidelines for Good Clinical Practice (GCP).

14.2 LAWS AND REGULATIONS

This Clinical Trial will be conducted in compliance with all international laws and regulations, and national laws and regulations of the country (ies) in which the Clinical Trial is performed, as well as any applicable guidelines.

14.3 INFORMED CONSENT

The Investigator (according to applicable regulatory requirements), or a person designated by the Investigator, and under the Investigator's responsibility, should fully inform the Patient of all pertinent aspects of the Clinical Trial including the written information giving approval/favorable opinion by the Ethics Committee (IRB/IEC). All participants should be informed to the fullest extent possible about the study, in language and terms they are able to understand.

Prior to a patient's participation in the Clinical Trial, the written Informed Consent Form should be signed, name filled in and personally dated by the patient or by the patient's legally acceptable representative, and by the person who conducted the informed consent discussion. A copy of the signed and dated written Informed Consent Form will be provided to the patient.

The Informed Consent Form used by the Investigator for obtaining the patient's informed consent must be reviewed and approved by the Sponsor prior to submission to the appropriate Ethics Committee (IRB/IEC) for approval/favorable opinion.

14.4 INSTITUTIONAL REVIEW BOARD/INDEPENDENT ETHICS COMMITTEE (IRB/IEC)

As required by local regulation, the Investigator or the Sponsor must submit this Clinical Trial Protocol to the appropriate Ethics Committee (IRB/IEC), and is required to forward to the respective other party a copy of the written and dated approval/favorable opinion signed by the Chairman with Ethics Committee (IRB/IEC) composition.

The Clinical Trial (study number, Clinical Trial Protocol title and version number), the documents reviewed (Clinical Trial Protocol, Informed Consent Form, Investigator's Brochure, Investigator's CV, etc.) and the date of the review should be clearly stated on the written (IRB/IEC) approval/favorable opinion.

Investigational Product will not be released at the study site and the Investigator will not start the study before the written and dated approval/favorable opinion is received by the Investigator and the Sponsor.

During the Clinical Trial, any amendment or modification to the Clinical Trial Protocol should be submitted to the Ethics Committee (IRB/IEC) before implementation, unless the change is necessary to eliminate an immediate hazard to the patients, in which case the IRB/IEC should be informed as soon as possible. It should also be informed of any event likely to affect the safety of patients or the continued conduct of the Clinical Trial, in particular any change in safety. All updates to the Investigator's Brochure will be sent to the Ethics Committee (IRB/IEC).

If requested, a progress report is sent to the Ethics Committee (IRB/IEC) at least annually and a summary of the Clinical Trial's outcome at the end of the Clinical Trial.

15 STUDY MONITORING

15.1 RESPONSIBILITIES OF THE INVESTIGATOR(S)

The Investigator(s) undertake(s) to perform the Clinical Trial in accordance with this Clinical Trial Protocol, ICH guidelines for Good Clinical Practice and the applicable regulatory requirements.

The Investigator is required to ensure compliance with all procedures required by the Clinical Trial Protocol and with all study procedures provided by the Sponsor (including security rules). The Investigator agrees to provide reliable data and all information requested by the Clinical Trial Protocol (with the help of the Case Report Form [CRF], Discrepancy Resolution Form [DRF] or other appropriate instrument) in an accurate manner according to the instructions provided and to ensure direct access to source documents by Sponsor representatives.

The Investigator may appoint such other individuals, as he/she may deem appropriate as Sub-Investigators to assist in the conduct of the Clinical Trial in accordance with the Clinical Trial Protocol. All Sub-Investigators shall be appointed and listed in a timely manner. The Sub-Investigators will be supervised by and work under the responsibility of the Investigator. The Investigator will provide them with a copy of the Clinical Trial Protocol and all necessary information.

15.2 RESPONSIBILITIES OF THE SPONSOR

The Sponsor of this Clinical Trial is responsible to Health Authorities for taking all reasonable steps to ensure the proper conduct of the Clinical Trial Protocol as regards ethics, Clinical Trial Protocol compliance, and integrity and validity of the data recorded on the Case Report Forms. Thus, the main duty of the Monitoring Team is to help the Investigator and the Sponsor maintain a high level of ethical, scientific, technical and regulatory quality in all aspects of the Clinical Trial.

At regular intervals during the Clinical Trial, the site will be contacted, through monitoring visits, letters or telephone calls, by a representative of the Monitoring Team to review study progress, Investigator and patient compliance with Clinical Trial Protocol requirements and any emergent problems. These monitoring visits, will include but not be limited to review of the following aspects : patient informed consent, patient recruitment and follow-up, Serious Adverse Event documentation and reporting, AE documentation, Investigational Product allocation, patient compliance with the Investigational Product regimen, Investigational Product accountability, concomitant therapy use and quality of data.

15.3 SOURCE DOCUMENT REQUIREMENTS

According to the ICH guidelines for Good Clinical Practice, the Monitoring Team must check the Case Report Form entries against the source documents, except for the pre-identified source data

directly recorded in the Case Report Form. The Informed Consent Form will include a statement by which the patient allows the Sponsor's duly authorized personnel, the Ethics Committee (IRB/IEC), and the regulatory authorities to have direct access to original medical records which support the data on the Case Report Forms (e.g., patient's medical file, appointment books, original laboratory records, etc.). These personnel, bound by professional secrecy, must maintain the confidentiality of all personal identity or personal medical information (according to confidentiality rules).

15.4 USE AND COMPLETION OF CASE REPORT FORMS (CRFS) AND ADDITIONAL REQUEST

It is the responsibility of the Investigator to maintain adequate and accurate CRFs (according to the technology used) designed by the Sponsor to record (according to Sponsor instructions) all observations and other data pertinent to the clinical investigation.

In this study an electronic CRF will be used to collect part of the requested information for all enrolled patients. All CRFs should be completed in their entirety to ensure accurate interpretation of data.

The computerized handling of the data by the Sponsor after receipt of the CRFs may generate additional requests (DRF) to which the Investigator is obliged to respond by confirming or modifying the data questioned.

A separate manual will describe in detail the procedures for using the electronic CRF. The Sponsor is responsible for ensuring that appropriate material is available at the investigative sites for completion of the electronic CRFs or for providing this material, if needed.

15.5 USE OF COMPUTERIZED SYSTEMS

Procedures shall be employed and controls designed to ensure the confidentiality of electronic records. Such procedures and controls shall include validation of systems to ensure accuracy and reliability, ability to generate accurate and complete copies of records, protection of records to enable retrieval, use of secure, computer-generated, time-stamped entries, use of operational system checks, use of device checks to determine validity of source data input, determination that person who develop, maintain, or use such systems have adequate education and training, the establishment and adherence of written policies to deter record falsification, the use of appropriate controls over systems documentation including the distribution of or use of documentation for system operation and maintenance, and revision and change control procedures which document time-sequenced development and modifications of systems documentation.

16 ADMINISTRATIVE RULES

16.1 CURRICULUM VITAE

A current copy in English of the curriculum vitae limited to the experience, qualification and training of each Investigator and Sub-Investigator will be provided to the Sponsor prior to the beginning of the Clinical Trial.

16.2 RECORD RETENTION IN STUDY SITES (S)

The Investigator must maintain confidential all study documentation, and take measures to prevent accidental or premature destruction of these documents.

It is recommended that the Investigator retain the study documents at least fifteen (15) years after the completion or discontinuation of the Clinical Trial.

However, applicable regulatory requirements should be taken into account in the event that a longer period is required.

The Investigator must notify the Sponsor prior to destroying any study essential documents following the Clinical Trial completion or discontinuation.

If the Investigator's personal situation is such that archiving can no longer be ensured by him/her, the Investigator shall inform the Sponsor and the relevant records shall be transferred to a mutually agreed upon designee.

17 CONFIDENTIALITY

All information disclosed or provided by the Sponsor (or any company/institution acting on their behalf), or produced during the Clinical Trial, including, but not limited to, the Clinical Trial Protocol, the CRFs, the Investigator's Brochure and the results obtained during the course of the Clinical Trial, is confidential. The Investigator and any person under his/her authority agrees to undertake to keep confidential and not to disclose the information to any third party without the prior written approval of the Sponsor.

However, the submission of this Clinical Trial Protocol and other necessary documentation to the Ethics Committee (IRB/IEC) is expressly permitted, the IRB/IEC members having the same obligation of confidentiality.

The Sub-Investigators shall be bound by the same obligation as the Investigator. The Investigator shall inform the Sub-Investigators of the confidential nature of the Clinical Trial.

The Investigator and the Sub-Investigators shall use the information solely for the purposes of the Clinical Trial, to the exclusion of any use for their own or for a third party's account.

18 PROPERTY RIGHTS

All information, documents and Investigational Product provided by the Sponsor or its designee are and remain the sole property of the Sponsor.

The Investigator shall not mention any information or the Product in any application for a patent or for any other intellectual property rights.

All the results, data, documents and inventions, which arise directly or indirectly from the Clinical Trial in any form, shall be the immediate and exclusive property of the Sponsor.

The Sponsor may use or exploit all the results at its own discretion, without any limitation to its property right (territory, field, continuance). The Sponsor shall be under no obligation to patent, develop, market or otherwise use the results of the Clinical Trial.

As the case may be, the Investigator and/or the Sub-Investigators shall provide all assistance required by the Sponsor, at the Sponsor's expense, for obtaining and defending any patent, including signature of legal documents.

19 DATA PROTECTION

- The patient's personal data and Investigator's personal data which may be included in the Sponsor database shall be treated in compliance with all applicable laws and regulations;
- When archiving or processing personal data pertaining to the Investigator and/or to the patients, the Sponsor shall take all appropriate measures to safeguard and prevent access to this data by any unauthorized third party.

20 INSURANCE COMPENSATION

The Sponsor certifies that it has taken out a liability insurance policy which covers the liability of the Investigator. This insurance policy is in accordance with local laws and requirements.

An insurance certificate will be provided to the Investigator in countries requiring this document.

The insurance of the Sponsor does not relieve the Investigator and the collaborators of any obligation to maintain their own liability insurance policy as required by applicable law.

21 SPONSOR AUDITS AND INSPECTIONS BY REGULATORY AGENCIES

For the purpose of ensuring compliance with the Clinical Trial Protocol, Good Clinical Practice and applicable regulatory requirements, the Investigator should permit auditing by or on the behalf of the Sponsor and inspection by regulatory authorities.

The Investigator agrees to allow the auditors/inspectors to have direct access to his/her study records for review, being understood that this personnel is bound by professional secrecy, and as such will not disclose any personal identity or personal medical information.

The Investigator will make every effort to help with the performance of the audits and inspections, giving access to all necessary facilities, data, and documents.

As soon as the Investigator is notified of a planned inspection by the authorities, he will inform the Sponsor and authorize the Sponsor to participate in this inspection.

The confidentiality of the data verified and the protection of the patients should be respected during these inspections.

Any result and information arising from the inspections by the regulatory authorities will be immediately communicated by the Investigator to the Sponsor.

The Investigator shall take appropriate measures required by the Sponsor to take corrective actions for all problems found during the audit or inspections.

22 PREMATURE DISCONTINUATION OF THE STUDY OR PREMATURE CLOSE-OUT OF A SITE

22.1 DECIDED BY THE SPONSOR IN THE FOLLOWING CASES:

- If the information on the product leads to doubt as to the benefit/risk ratio;
- If the Investigator has received from the Sponsor all Investigational Product, means and information necessary to perform the Clinical Trial and has not included any patient after a reasonable period of time mutually agreed upon;
- In the event of breach by the Investigator of a fundamental obligation under this agreement, including but not limited to breach of the Clinical Trial Protocol, breach of the applicable laws and regulations or breach of the ICH guidelines for Good Clinical Practice;
- If the total number of patients are included earlier than expected;

In any case the Sponsor will notify the Investigator of its decision by written notice.

22.2 DECIDED BY THE INVESTIGATOR

The Investigator must notify (30 days' prior notice) the Sponsor of his/her decision and give the reason in writing.

In all cases (decided by the Sponsor or by the Investigator), the appropriate Ethics Committee(s) (IRB/IEC) and Health Authorities should be informed according to applicable regulatory requirements.

23 CLINICAL TRIAL RESULTS

- The Sponsor will be responsible for preparing a Clinical Study Report.
- When the data from all investigational sites have been fully analyzed by the Sponsor, the latter will communicate the results of the Clinical Trial to the Investigator(s).

24 PUBLICATIONS AND COMMUNICATIONS

The Investigator undertakes not to make any publication or release pertaining to the Study and/or results of the Study the Sponsor's prior written consent, being understood that the Sponsor will not unreasonably withhold its approval.

If the Study is being conducted at multiple sites, the Sponsor agrees that, consistent with scientific standards, first presentation or publication of the results of the Study shall be made only as part of a publication of the results obtained by all sites performing the Protocol. However, if no multicenter publication has occurred within twelve (12) months of the completion of this Study at all sites, the Investigator shall have the right to publish or present independently the results of this Study patient to the review procedure set forth herein.

The Investigator shall provide the Sponsor with a copy of any such presentation or publication derived from the Study for review and comment at least thirty (30) days in advance of any presentation or submission for publication. In addition, if requested by the Sponsor, any presentation or submission for publication shall be delayed for a limited time, not to exceed ninety (90) days, to allow for filing of a patent application or such other measures as the Sponsor deems appropriate to establish and preserve its proprietary rights.

The Investigator shall not use the name(s) of the Sponsor and/or its employees in advertising or promotional material or publication without the prior written consent of the Sponsor. The Sponsor shall not use the name(s) of the Investigator and/or the collaborators in advertising or promotional material or publication without having received his/her and/or their prior written consent(s).

The Sponsor has the right at any time to publish the results of the study.

For the main publication, authors' ranking will be based on their contribution to design and management of the study and to the number of eligible patients included by centers (only 1 author per center) and the principal investigator will be at least the last author. Other investigators will be acknowledged in the publication

25 CLINICAL TRIAL PROTOCOL AMENDMENTS

All appendices attached hereto and referred to herein are made part of this Clinical Trial Protocol.

The Investigator should not implement any deviation from, or changes of the Clinical Trial Protocol without agreement by the Sponsor and prior review and documented approval/favorable opinion from the IRB/IEC of an amendment, except where necessary to eliminate an immediate hazard(s) to Clinical Trial Patients, or when the change(s) involves only logistical or administrative aspects of the trial. Any change agreed upon will be recorded in writing, the written amendment will be signed by the Investigator and by the Sponsor and the signed amendment will be filed with this Clinical Trial Protocol.

Any amendment to the Clinical Trial Protocol requires written approval/favorable opinion by the Ethics Committee (IRB/IEC) prior to its implementation, unless there are overriding safety reasons.

In some instances, an amendment may require a change to the Informed Consent Form. The Investigator must receive an IRB/IEC approval/favorable opinion concerning the revised Informed Consent Form prior to implementation of the change.

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27 APPENDICES

Appendix A Cockcroft-Gault Formula

Cockcroft-Gault Formula [29]

$$\text{Creatinine Clearance} = \frac{(140 - \text{Age [yrs]}) \times \text{Body Mass (kg)}}{\text{Plasma Creatinine (mg/dL)} \times 72} \times \text{Gender Correction Factor} \\ (\text{male : 1.00; female : 0.85})$$

Appendix B Response evaluation criteria in solid tumors (RECIST) quick reference

Eligibility

- Only patients with measurable disease at baseline should be included in protocols where objective tumor response is the primary endpoint.

Measurable disease - the presence of at least one measurable lesion. If the measurable disease is restricted to a solitary lesion, its neoplastic nature should be confirmed by cytology/histology.

Measurable lesions - lesions that can be accurately measured in at least one dimension with longest diameter ≥ 20 mm using conventional techniques or ≥ 10 mm with spiral CT scan.

Non-measurable lesions - all other lesions, including small lesions (longest diameter < 20 mm with conventional techniques or < 10 mm with spiral CT scan), i.e., bone lesions, leptomeningeal disease, ascites, pleural/pericardial effusion, inflammatory breast disease, lymphangitis cutis/pulmonis, cystic lesions, and also abdominal masses that are not confirmed and followed by imaging techniques; and

- All measurements should be taken and recorded in metric notation, using a ruler or calipers. All baseline evaluations should be performed as closely as possible to the beginning of treatment and never more than 4 weeks before the beginning of the treatment.
- The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up.
- Clinical lesions will only be considered measurable when they are superficial (e.g., skin nodules and palpable lymph nodes). For the case of skin lesions, documentation by color photography, including a ruler to estimate the size of the lesion, is recommended.

Methods of Measurement

- CT and MRI are the best currently available and reproducible methods to measure target lesions selected for response assessment. Conventional CT and MRI should be performed with cuts of 10 mm or less in slice thickness contiguously. Spiral CT should be performed using a 5 mm contiguous reconstruction algorithm. This applies to tumors of the chest, abdomen and pelvis. Head and neck tumors and those of extremities usually require specific protocols.
- Lesions on chest X-ray are acceptable as measurable lesions when they are clearly defined and surrounded by aerated lung. However, CT is preferable.
- When the primary endpoint of the study is objective response evaluation, ultrasound (US) should not be used to measure tumor lesions. It is, however, a possible alternative to clinical measurements of superficial palpable lymph nodes, subcutaneous lesions and thyroid nodules. US might also be useful to confirm the complete disappearance of superficial lesions usually assessed by clinical examination.

- The utilization of endoscopy and laparoscopy for objective tumor evaluation has not yet been fully and widely validated. Their uses in this specific context require sophisticated equipment and a high level of expertise that may only be available in some centers. Therefore, the utilization of such techniques for objective tumor response should be restricted to validation purposes in specialized centers. However, such techniques can be useful in confirming complete pathological response when biopsies are obtained.
- Tumor markers alone cannot be used to assess response. If markers are initially above the upper normal limit, they must normalize for a patient to be considered in complete clinical response when all lesions have disappeared.
- Cytology and histology can be used to differentiate between PR and CR in rare cases (e.g., after treatment to differentiate between residual benign lesions and residual malignant lesions in tumor types such as germ cell tumors).

Baseline documentation of “Target” and “Non-Target” lesions

- All measurable lesions up to a maximum of five lesions per organ and 10 lesions in total, representative of all involved organs should be identified as target lesions and recorded and measured at baseline.
- Target lesions should be selected on the basis of their size (lesions with the longest diameter) and their suitability for accurate repeated measurements (either by imaging techniques or clinically).
- A sum of the longest diameter (LD) for all target lesions will be calculated and reported as the baseline sum LD. The baseline sum LD will be used as reference by which to characterize the objective tumor.
- All other lesions (or sites of disease) should be identified as non-target lesions and should also be recorded at baseline. Measurements of these lesions are not required, but the presence or absence of each should be noted throughout follow-up.

Response Criteria

Evaluation of target lesions	
Complete Response (CR)	Disappearance of all target lesions
Partial Response (PR)	At least a 30% decrease in the sum of the LD of target lesions, taking as reference the baseline sum LD
Progressive Disease (PD)	At least a 20% increase in the sum of the LD of target lesions, taking as reference the smallest sum LD recorded since the treatment started or the appearance of one or more new lesions
Stable disease (SD)	Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum LD since the treatment started

Evaluation of non-target lesions	
Complete Response (CR)	Disappearance of all non-target lesions and normalization of tumor marker level
Incomplete Response / Stable Disease (SD)	Persistence of one or more non-target lesion(s) or/and maintenance of tumor marker level above the normal limits
Progressive Disease (PD)	Appearance of one or more new lesions and/or unequivocal progression of existing non-target lesions ¹

¹ Although a clear progression of “non target” lesions only is exceptional, in such circumstances, the opinion of the treating physician. Should prevail and the progression status should be confirmed later on by the review panel (or study chairmen).

Evaluation of best overall response

The best overall response is the best response recorded from the start of the treatment until disease progression/recurrence (taking as reference for PD the smallest measurements recorded since the treatment started). In general, the patient’s best response assignment will depend on the achievement of both measurement and confirmation criteria.

Target lesions	Non-target lesions	New lesions	Overall response
CR	CR	No	CR
CR	Incomplete response / SD	No	PR
PR	Non-PD	No	PR
SD	Non-PD	No	SD
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

- Patients with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be classified as having “symptomatic deterioration”. Every effort should be made to document the objective progression even after discontinuation of treatment.
- In some circumstances it may be difficult to distinguish residual disease from normal tissue. When the evaluation of complete response depends on this determination, it is recommended that the residual lesion be investigated (fine needle aspirate/biopsy) to confirm the complete response status.

Confirmation

- The main goal of confirmation of objective response is to avoid overestimating the response rate observed. In cases where confirmation of response is not feasible, it should be made clear when reporting the outcome of such studies that the responses are not confirmed.

- To be assigned a status of PR or CR, changes in tumor measurements must be confirmed by repeat assessments that should be performed no less than 4 weeks after the criteria for response are first met. Longer intervals as determined by the study protocol may also be appropriate.
- In the case of SD, follow-up measurements must have met the SD criteria at least once after study entry at a minimum interval (in general, not less than 6-8 weeks) that is defined in the study protocol

Duration of overall response

- The duration of overall response is measured from the time measurement criteria are met for CR or PR (whichever status is recorded first) until the first date that recurrence or PD is objectively documented, taking as reference for PD the smallest measurements recorded since the treatment started.

Duration of stable disease

- SD is measured from the start of the treatment until the criteria for disease progression are met, taking as reference the smallest measurements recorded since the treatment started.
- The clinical relevance of the duration of SD varies for different tumor types and grades. Therefore, it is highly recommended that the protocol specify the minimal time interval required between two measurements for determination of SD. This time interval should take into account the expected clinical benefit that such a status may bring to the population under study.

Response review

- For trials where the response rate is the primary endpoint it is strongly recommended that all responses be reviewed by an expert(s) independent of the study at the study's completion. Simultaneous review of the patients' files and radiological images is the best approach.

Reporting of results

- All patients included in the study must be assessed for response to treatment, even if there are major protocol treatment deviations or if they are ineligible. Each patient will be assigned one of the following categories: 1) complete response, 2) partial response, 3) stable disease, 4) progressive disease, 5) early death from malignant disease, 6) early death from toxicity, 7) early death because of other cause, or 9) unknown (not assessable, insufficient data).
- All of the patients who met the eligibility criteria should be included in the main analysis of the response rate. Patients in response categories 4-9 should be considered as failing to respond to treatment (disease progression). Thus, an incorrect treatment schedule or drug administration does not result in exclusion from the analysis of the response rate. Precise definitions for categories 4-9 will be protocol specific.
- All conclusions should be based on all eligible patients.
- Subanalyses may then be performed on the basis of a subset of patients, excluding those for whom major protocol deviations have been identified (e.g., early death due to other reasons,

early discontinuation of treatment, major protocol violations, etc.). However, these subanalyses may not serve as the basis for drawing conclusions concerning treatment efficacy, and the reasons for excluding patients from the analysis should be clearly reported.

- The 95% confidence intervals should be provided.

Appendix C Specialized blood sample collection, handling and shipping

Recommended methods for pharmacokinetic, and anti-aflibercept antibody blood sample collection and handling

Since study treatment is administered intravenously, blood samples for pharmacokinetic analysis should not be collected from the arm where study treatment is infused, or from a central line used for study treatment administration. For samples collected through a catheter, 1 mL of blood should be withdrawn and discarded at each sampling time to ensure that the solution used to maintain catheter patency does not dilute the blood sample.

It is extremely important to collect all blood samples as close to the protocol-specified times as possible. The reasons for any missed or lost blood samples should be documented. Both the scheduled and actual dates/times of blood collection should appear on the blood collection record at the study site. The times of aflibercept or placebo administration should also be precisely recorded.

Blood collection for aflibercept pharmacokinetic analysis in CTAD plasma

At the protocol-specified intervals indicated in Section 9.3, "CTAD" vacutainer tubes (containing 1 mL of citrate buffer, sodium citrate, and 4.2 mg of citric acid) will be used to collect 4 mL of whole blood from patients for plasma preparation for the determination of circulating free aflibercept and VEGF: aflibercept complex levels for pharmacokinetic analysis.

Because platelet lysis can release VEGF into the serum and possibly affect aflibercept measurements, it is important to carefully follow the steps outlined below:

1. Draw blood slowly, using the largest-bore catheter that is feasible for the patient and transfer blood to a CTAD vacutainer tube (B-D Hemogard, sedimentation rate determination 4.5 mL tubes), being careful to minimize agitation of the sample.
2. gently mix tube by inverting 4 times.
3. Centrifuge at 2000 x g for 15 minutes at room temperature within 1 hour of blood draw.
4. Within 30 minutes after centrifugation, draw off plasma very slowly with transfer pipette to within 0.5 cm of the buffy coat, taking great care not to disturb the buffy coat (any contamination may invalidate the assay).
5. Pipette the plasma specimen into two separate plastic cryovials.
6. Complete corresponding lines for each blood draw on the shipping log. Include patient identification number, initials, and date of specimen collection.
7. All specimens must be stored frozen at -20°C (or colder) in a freezer that is not frost-free until shipped to Regeneron.
8. The Regeneron Specimen Log Sheets must be completed and included along with all shipments to Regeneron.

9. Only the primary samples of each patient should be sent with the first shipment to ensure that duplicates are available at the site for backup in case of damage during shipment.
10. On the day specimens will be shipped, please notify Regeneron to expect arrival.

Sufficient dry ice is to be included for samples to remain frozen for at least 48 hours (recommend at least 14 lbs. of dry ice).

For free aflibercept and VEGF: aflibercept complex assays, the transport of plasma samples from the clinical site to Regeneron Pharmaceuticals, Inc. will take place in containers of solid CO₂ and be coordinated with:



Blood collection for Anti- aflibercept Antibody analysis

At the protocol-specified intervals indicated in Section 9.3, red-top vacutainer tubes without clot activator will be used to collect 4 mL of whole blood from patients for serum preparation for the analysis of anti- aflibercept antibodies formation. The following procedures should be followed:

1. Complete the pre-printed labels with Investigator's name, patient identification numbers, initials and date of specimen collection.
2. Allow serum to clot for 30 minutes. Spin at 2000 x g for 15 minutes to separate clot from serum.
3. Pipette the serum specimen into two separate plastic cryovials (supplied by Regeneron Pharmaceuticals, Inc.).
4. Complete corresponding lines for each blood draw on the shipping log. Include: patient identification number, initials, date of specimen collection.
5. All specimens must be stored frozen at -20°C (or colder) until shipped to the laboratory designated by Regeneron Pharmaceuticals, Inc.
6. The Specimen Log Sheet must be completed and included along with all shipments.
7. Only the primary samples of each patient should be sent with the first shipment. This is to ensure that in case of damage during shipment there are duplicates at the site for backup.
8. On the day you will be shipping specimens, please notify [REDACTED] or designee at Regeneron to expect arrival of the package.

Sufficient dry ice is to be included for samples to remain frozen for at least 48 hours (recommend 14 lbs. of dry ice).

For the analysis of anti- aflibercept antibodies, the transport of serum samples from the clinical site to Regeneron Pharmaceuticals, Inc. will take place in containers of solid CO2 and be coordinated with:

[REDACTED]

Blood collection for the assay of free endogenous VEGF

Blood samples (4 mL) will be collected in **CPD (or ACD) tubes** and centrifuged at +4°C, at 2000 g for 20 minutes. The plasma will be transferred into two polypropylene screw caps tubes (500 µL in each tube as a minimum) and frozen immediately at -20°C in the upright position within 1 hour after the puncture.

All specimens must be stored frozen at -20°C until shipped to sanofi-aventis

The Specimen Log Sheet must be completed and included along with all shipments.

One week before specimen shipment, please notify [REDACTED] to expect arrival of the package.

The plasma free VEGF concentrations will be determined at sanofi-aventis, GMPK Alfortville under the responsibility of [REDACTED] at the following address:

Sanofi-aventis Recherche and Développement

[REDACTED]

Appendix D ECOG performance status scale

Grade	Description
0	Normal activity. Fully active, able to carry on all pre-disease performance without restriction.
1	Symptoms, but ambulatory. Restricted in physically strenuous activity, but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework, office work).
2	In bed < 50% of the time. Ambulatory and capable of all self-care, but unable to carry out any work activities. Up and about more than 50% waking hours.
3	In bed > 50% of the time. Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.
4	100% bedridden. Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
5	Dead

Appendix E National Cancer Institute Common Terminology Criteria for Adverse Events Version 3.0

When 2 criteria are available for similar toxicities, the one resulting in the more severe grade should be used.

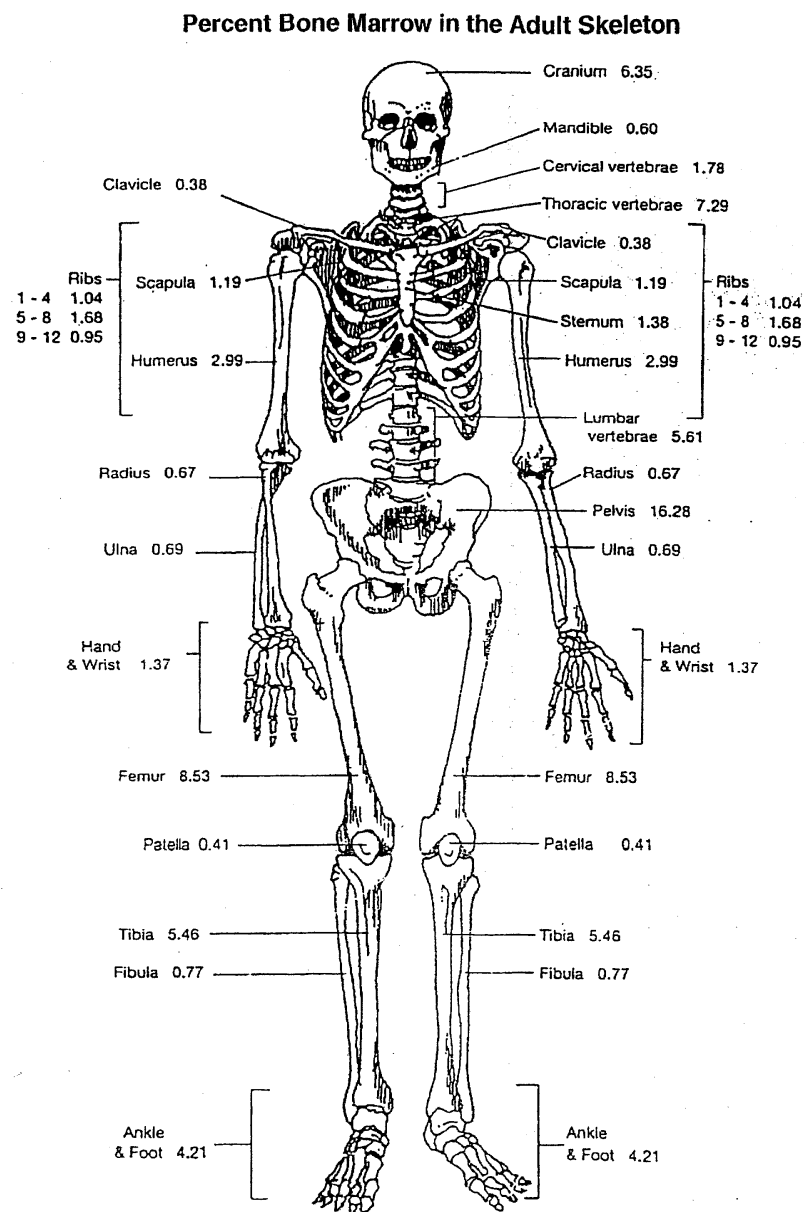
If the toxicity causes death, then it is Grade 5.

The evaluator must attempt to discriminate between disease / treatment and related signs / symptoms.

An accurate baseline prior to therapy is essential.

See [25] Common Terminology Criteria for Adverse Events (CTCAE) v3.0 March 31, 2003, Publish Date: August 9, 2006 (http://ctep.info.nih.gov/reporting/ctc_v30.html).

Appendix F Percent bone marrow in the human skeleton



Woodward Holodny E. A summary of the data of Mechanik on the distribution of human bone marrow. Phys Med Biol. 1960;5:57-59

Appendix G Lung Cancer Symptom Scale (LCSS)

Directions: Please place a mark along each line where it would best describe the symptoms of your lung cancer DURING THE PAST DAY (within the last 24 hours).

HOW IS THE WEATHER TODAY?

AS GOOD AS IT	_____	AS BAD AS IT
COULD BE	_____	COULD BE

1 HOW IS YOUR APPETITE?

AS GOOD AS IT	_____	AS BAD AS IT
COULD BE	_____	COULD BE

2 HOW MUCH FATIGUE DO YOU HAVE?

NONE	_____	AS MUCH AS IT COULD
	_____	BE

3 HOW MUCH COUGHING DO YOU HAVE?

NONE	_____	AS MUCH AS IT COULD
	_____	BE

4 HOW MUCH SHORTNESS OF BREATH DO YOU HAVE?

NONE	_____	AS MUCH AS IT COULD
	_____	BE

5 HOW MUCH BLOOD DO YOU SEE IN YOUR SPUTUM?

NONE	_____	AS MUCH AS IT COULD
	_____	BE

6 HOW MUCH PAIN DO YOU HAVE?

NONE	_____	AS MUCH AS IT
	_____	COULD BE

7 HOW BAD ARE YOUR SYMPTOMS FROM LUNG CANCER?

	_____	AS BAD AS THEY
I HAVE NONE	_____	COULD BE

8 HOW MUCH HAS YOUR ILLNESS AFFECTED YOUR ABILITY TO

CARRY OUT NORMAL ACTIVITIES?

NOT AT ALL | _____ | SO MUCH THAT I CAN
| _____ | DO NOTHING FOR MYSELF

9 HOW WOULD YOU RATE THE QUALITY OF YOUR LIFE TODAY?

_____ |
VERY HIGH | _____ | VERY LOW

Appendix H New York Heart Association Classification

Functional Class	Description
0	No cardiac disease or limitations
I	Patients with cardiac disease but without limitations of physical activity. Ordinary physical activity does not cause undue fatigue, dyspnea or palpitation.
II	Patients with cardiac disease that results in slight limitations of physical activity. Ordinary physical activity results in fatigue, palpitation, dyspnea or angina.
III	Patients with cardiac disease that results in marked limitation of physical activity. Although patients are comfortable at rest, less than ordinary activity will lead to symptoms.
IV	Patients with cardiac disease that results in an inability to carry on physical activity without discomfort. Symptoms of congestive heart failure are present even at rest. With any physical activity, increased discomfort is experienced.